



ADBI Working Paper Series

**RENMINBI INVOICING UNDER
DOLLAR DOMINANCE:
EVIDENCE ON INERTIA,
INSTITUTIONS, AND POLICY**

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Abstract

This paper examines the conditions under which renminbi invoicing has expanded despite the continued dominance of the United States (US) dollar in international trade. Using IMF trade invoicing data for 70 economies from 2016 to 2023, as reported by Boz et al. (2025), we show that RMB invoicing is systematically shaped by trade dependence on the People's Republic of China (PRC), exchange rate regimes, capital and foreign exchange controls, settlement infrastructure, and the strength of political and institutional ties with the PRC. While dollar-oriented exchange rate regimes and capital liberalization tend to restrict use of the renminbi, renminbi invoicing is expanding in economies with remaining controls and deeper integration into PRC-led financial, logistical, and digital networks, indicating that RMB internationalization is a gradual, complementary process rather than a substitute for the dollar.

Keywords: Dominant Currency Paradigm, internationalization of the renminbi, invoice currency, foreign exchange controls, logistics

JEL Classification: F23, F31, F33

Contents

1.	INTRODUCTION	1
2.	RECENT RESEARCH ON INVOICE CURRENCY CHOICE.....	3
3.	EXAMINATION AND ANALYSIS OF EXPLANATORY VARIABLES RELATED TO INVOICE CURRENCY CHOICE.....	8
3.1	Exchange Rates and Their Institutional Factors	8
3.2	Examination of Capital Market Factors.....	10
3.3	The PRC's Policies: Currency Swaps and Overseas Financial Institutions (Financial Infrastructure)	16
3.4	Other PRC Government-Led Policies: Number of Visits by PRC Government Officials and Their Counterparts, and Other Related Policies ..	21
4.	EMPIRICAL ANALYSIS OF FACTORS PROMOTING RMB-INVOICED TRANSACTIONS	29
4.1	Examination of Dependent Variables	29
4.2	Empirical Analysis Model	31
4.3	Empirical Analysis Results: Share of RMB-invoiced Imports	33
4.4	Empirical Analysis Results: Share of RMB-Invoiced Exports.....	35
4.5	Discussion	37
5.	CONCLUSION AND FUTURE RESEARCH	39
	REFERENCES	41

1. INTRODUCTION

Since the late 2010s, the environment surrounding currency choice in international trade and cross-border investment has been undergoing a major transformation. Although there has been no major change in the United States (US) dollar's dominant position in global trade, movements to reevaluate excessive reliance on the US dollar have become apparent in various countries against the backdrop of rising geopolitical and political-economic risks, such as the prolonged conflict between the US and the People's Republic of China (PRC), the normalization of financial sanctions triggered by the Russian invasion of Ukraine, and the recent resurgence of trade friction through tariff policies. Policy initiatives aimed at promoting so-called "local currency transactions"—where trade and investment are conducted either in the domestic currency or in that of the counterparty—are being advanced, primarily in emerging economies and Global South countries.

Rather than being seen as a radical challenge to the existing international monetary system, these moves should be understood through the lens of practical risk management and economic security. While US dollar invoicing still holds an overwhelming advantage in terms of liquidity, inertia, and network effects, it is also susceptible to political influence through sanctions and financial regulations. Particularly for countries that have faced the risk of restricted access to international payment infrastructure and major banking networks, currency diversification is increasingly seen not merely as a cost-cutting measure but as an institutional option for maintaining stable external transactions. From a geopolitical perspective, countries located far from the United States (most of which are emerging markets or developing nations) tend to use the US dollar as their invoicing currency and remain dependent on it, though data indicate that this dependence is decreasing in some major economies. Since 2021, the correlation between the use of invoicing currencies and geopolitical distance from the currency's issuing country has trended downward, suggesting that geopolitical fragmentation is progressing. Since the Russian invasion of Ukraine, the influence of geopolitical distance on the choice of invoicing currency has become pronounced. In countries far from the US and the eurozone, the use of the dollar and the euro has declined, with greater reliance on the renminbi, local currencies, and other third-country currencies. Conversely, in countries that have become more distant from the PRC, there has been a tendency to reduce the use of domestic and third-country currencies in favor of the US dollar.¹

Previous research, primarily conducted by the IMF, has empirically demonstrated the phenomenon known as the "Dominant Currency Paradigm" (DCP), in which the dominant position of the US dollar exerts a strong influence on corporate pricing and the adjustment of trade volumes. In recent years, however, attention has focused on the fact that, even as the DCP persists, the use of nondollar currencies is expanding in certain regions and sectors. The PRC's efforts to internationalize the renminbi have steadily expanded its use in trade settlements and certain capital transactions through government-led institutional reforms, such as currency swap agreements, the establishment of clearing banks, and the development of the Cross-border Interbank Payment System (CIPS). Furthermore, in ASEAN countries, alongside the expansion of intraregional trade, the use of regional currencies is progressing through frameworks such as the Local Currency Settlement Framework (LCSF). In specific countries and

¹ At the Autumn Conference of the Japanese Society of Finance (15 November 2025), Yosuke Tsuyuguchi of Teikyo University, citing this paper, pointed out that geopolitical factors do not necessarily lead to dedollarization.

regions, although their share in actual trade statistics remains small, the presence of currencies other than the US dollar is becoming clear.

Furthermore, as the world's largest energy importer, the PRC has promoted the use of the renminbi in transactions with resource-exporting countries, with an increasing number of renminbi settlements in oil and gas imports from suppliers such as the Russian Federation and Saudi Arabia. Such a shift in currency choice for resource and energy transactions—driven by developments such as the Russian invasion of Ukraine and the expansion of bilateral settlements in non-US dollar currencies—indicates that dedollarization is increasingly supported by transaction-based demand, particularly among emerging economies. The growth of local-currency invoicing, especially for import payments, underscores the importance of exchange rate risk allocation between exporters and importers and highlights the role of domestic factors, including exchange control regimes, financial market development, and external balance sheet structures, in shaping currency choice.

The purpose of this paper is to conduct an empirical analysis of the conditions under which renminbi-invoiced transactions expand, based on factors derived from a detailed examination of published statistics and institutional frameworks across various countries, primarily in Asia, as conducted by Tomizawa and Shimizu (2026). Tomizawa and Shimizu (2026), based on data published by governments and central banks in countries other than the PRC, noted that transactions denominated in local currencies and the renminbi have been increasing in countries such as Thailand and Brazil. They highlighted geopolitical factors that had not previously been identified in existing research as drivers of local currency use, the implementation of policies promoting such use, technological factors such as the movement of people and the spread of digital payments, and the construction of payment infrastructure—all of which may contribute to the expansion of renminbi usage. The contribution of this study lies in the fact that, by cross-referencing the mechanisms promoting local currency transactions and the results of factor analysis revealed through the analysis of local currency-denominated transactions in various countries and regions, we extracted new explanatory variable data from various statistics and publicly available information, constructed a dataset, and identified a new model using various “PRC policy factor” variables—which had not been employed in previous studies—as explanatory variables.

Furthermore, the finding of significance for many of these variables constitutes a key contribution of this study. Specifically, using the share of renminbi-denominated invoices in each country's exports, as published by Boz et al. (2025, IMF) as the dependent variable, and by attempting to quantify not only the factors indicating the internationalization of the renminbi used in previous studies but also new factors, we examine the roles played by factors such as currency and capital controls, financial infrastructure, and the formation of regional economic blocks. Through this, we aim to present a perspective that views the impact of recent changes in the global situation on the international monetary system not as a simple binary opposition of “dedollarization” but as a more multilayered, gradual structural change.

The structure of this paper is as follows. In Section 2, we summarize the key independent and dependent variables used in empirical analyses of currency choice and invoice currency in major prior studies, as well as the findings obtained. Next, in Section 3, we analyze phenomena that have not previously been identified as factors promoting local currency usage—specifically, those driving the increasing use of local currencies other than the US dollar—and attempt to quantify them as explanatory variables. Then, in Section 4, we conduct an empirical analysis using the share of the renminbi as the invoice currency in each country's exports and imports to the rest of

the world as the dependent variable, and the new explanatory variables discussed in Section 2. Finally, in Section 5, we summarize our conclusions and future research directions in discussing the results.

2. RECENT RESEARCH ON INVOICE CURRENCY CHOICE

This section summarizes the main findings of recent research on currency choice and invoice currencies, including the Chinese renminbi, as well as the explanatory variables used in these analyses.

Boz et al. (2025, IMF) analyze invoicing–currency patterns in global trade and the impact of geopolitical factors (Table 1). Specifically, they conduct an empirical analysis using the share of invoicing currencies (USD, EUR, RMB) in each country’s exports as the dependent variable, and the share of exports to the issuing countries of those currencies (the United States, the eurozone, and the PRC) as independent variables. Their main findings indicate that a limited number of currencies, including the US dollar and the euro, continue to dominate as invoicing currencies in global trade, and their shares remain stable even in the face of geopolitical factors. On the other hand, while the use of the renminbi has been steadily increasing since the early 2010s and expanding beyond Asia, its share remains relatively small.

Table 1: Boz et al. (IMF 2025)

Category	Contents
Dependent Variable	Share of invoice currencies in each country’s exports (USD/EUR/RMB)
Explanatory Variable	- Invoice currency share in the previous period (inertia)
	- Ratio of exports to the country issuing the invoicing currency
	- Ratio of exports to countries with currency pegs
	- Homogeneity of exported goods (ratio of fuels, ores, and agricultural raw materials; ratio of petroleum)
	- Bilateral exchange rates
	- Geopolitical distance (from the US/EU/PRC) (UN voting distance)
Key Findings	- The USD remains globally dominant
	- Use of the RMB is increasing but remains limited
	- Geopolitical divisions (especially since 2021) have a strong influence on the choice of invoicing currency
	- Countries with greater political distance from the US/EU are shifting toward the RMB and their own currencies
# of sample	132 countries
Analysis period	1990–2023 (invoice currency share estimates are primarily for 1999–2023)

Summary by the author.

Furthermore, from a geopolitical perspective, countries located far from the United States (most of which are emerging markets or developing countries) tend to use the US dollar as the invoicing currency and remain dependent on it, while the study indicates that this dependence is decreasing in some major economies. Since 2021, the correlation between the use of invoicing currencies and the geopolitical distance from the currency’s issuing country has trended downward, suggesting that geopolitical fragmentation is progressing. Since the Russian invasion of Ukraine, the influence of geopolitical distance on the choice of invoicing currency has become pronounced. In countries geographically distant from the United States and the eurozone, the use of the US dollar and the euro has declined, with a shift toward the Chinese yuan,

domestic currencies, and third-country currencies. Conversely, countries that have become more distant from the PRC have tended to reduce their use of domestic and third-country currencies in favor of the US dollar.

In addition, the paper analyzes the relationship between the share of exports of resources and primary commodities. While commodities such as oil are still frequently traded in US dollars, and although transactions in renminbi and other currencies have increased recently, the paper states that no systematic evidence has been found to substantiate the effectiveness of policy initiatives aimed at reducing dependence on the US dollar in oil exports.

The paper provides a dataset covering 132 countries from 1990 to 2023, showing the share of invoicing currencies for exports and imports, specifically the US dollar, the euro, the renminbi, and other currencies (such as domestic currencies not listed above).

Perez-Saiz and Zhang (2023, IMF) analyze the use of the renminbi as an international currency, regional patterns, and the role of local-currency swaps and offshore renminbi clearing banks, both of which are implemented by the People's Bank of China. They note that while the global use of the renminbi is still in its early stages, accounting for only 2% of global settlements, there are significant regional differences in usage, with its use expanding particularly in Asia and certain emerging market economies (such as Mongolia, the Lao People's Democratic Republic, Chile, Türkiye, and Argentina). Furthermore, their empirical analysis confirms that local-currency swaps and renminbi clearing banks are effective in promoting the use of the renminbi in international settlements. Furthermore, the study noted that political distance (measured using the correlation of countries' voting patterns at the UN General Assembly to gauge political distance from the PRC) and geographical distance are also key factors influencing the use of the renminbi. In particular, it indicated that the renminbi is spreading across Asia and that its role as a regional currency is likely to expand further.

Table 2: Perez-Saiz and Zhang (2023, IMF)

Category	Contents
Dependent Variable	Share of the RMB in international settlements with PRC (SWIFT MT data)
Explanatory Variable	- RMB usage in the previous period (inertia)
	- Swap lines (dummy)
	- Presence of RMB clearing banks (dummy)
	- Trade integration with PRC (import-export ratio)
	- Geographical distance (capital-PRC)
	- Political distance (UN voting distance)
	- Other (GDP, income, financial deepening, inflation, etc.)
Key Findings	- The effect of swap lines alone is unclear
	- Countries with deep trade ties to PRC expand RMB usage
	- Clearing banks contribute to promoting the use of RMB settlements
	- Political and geographical distance are also important factors
# of sample	Not explicitly stated in the document (using SWIFT data)
Analysis period	The year of the SWIFT data is not specified

Summary by the author.

Georgiadis et al. (2021, ECB) noted that approximately 40% of global trade is denominated in US dollars, far exceeding the US's share of global trade (approximately 10%). They demonstrated that strategic complementarity (the incentive to price in the

same currency as competitors) and integration into global value chains (GVCs) reinforce the use of the dollar, concluding that the dollar holds a dominant position as a “reserve currency” even in transactions with third countries.

On the other hand, with regard to the euro, the report noted that while euro-denominated transactions are common within, and in trade with, Europe, the euro plays a weaker role as a “vehicle currency” in transactions with third countries than the dollar, and that policy support from European authorities has been limited. In contrast, while the PRC’s expanding trade has strengthened the US dollar’s dominance and relatively diminished the euro’s status, the authors noted that the People’s Bank of China’s currency swap agreements are promoting renminbi-denominated transactions and that the renminbi’s internationalization relies more on policy than on fundamentals. The report demonstrated that the existence of currency swap agreements makes it easier for overseas financial institutions to raise funds and extend loans in renminbi, and for importers and exporters to choose renminbi for settlement. Furthermore, the agreements mitigate the risk of renminbi liquidity shortages, lower the barrier to companies selecting the renminbi as an invoicing currency, and increase the number of first-time users through network effects. Furthermore, the authors confirmed a trend of increasing renminbi-denominated trade invoices in countries that have signed swap agreements, as they showed that the greater a country’s trade share (dependence) with the PRC, the more renminbi invoices rise, with this trend being particularly strong for imports. They also pointed out that while the increase in renminbi invoices reduces the use of the US dollar and the euro, the impact on the US dollar is particularly significant.

Table 3: Georgiadis et al. (2021, ECB)

Category	Contents
Dependent Variable	Share of invoice currencies in a country’s exports and imports (USD/EUR/RMB/other)
Explanatory Variable	- Ratio of trading partners for exports and imports (US/EU)
	- Degree of integration into global value chains (GVCs)
	- Exchange rates (USD/EUR)
	- Country size, exchange rate regime, financial market development, etc. (captured via fixed effects)
Key Findings	- The USD dominance is maintained through strategic complementarity with competitors and GVC structures
	- While the euro is widely used within the EU, it lags behind the dollar globally
	- PRC’s swap agreements strongly boost RMB usage (having a greater impact than EU policies)
# of sample	102 countries (based on the Boz et al. dataset)
Analysis period	1999–2019

Summary by the author.

Furthermore, Ito and Chinn (2014), who focus on the rise of the renminbi in the 2010s and summarize its impact on RMB usage, analyzed factors such as the level of financial market development and capital account liberalization (KAOPEN index) that influence the choice of invoicing currency in international trade. Specifically, the study showed that the more developed a country’s financial markets are, the less likely it is to invoice in US dollars, and countries with open capital accounts tend to invoice in euros or their own currency. As regards the renminbi, while the actual renminbi invoicing ratio is below 10%, the model suggests that a 20%–30% ratio is appropriate, attributing the slow adoption to inertia.

Table 4: Ito and Chinn (2014, ADBI)

Category	Contents
Dependent Variable	Share of export invoice currencies (USD/EUR/local currency)
Explanatory Variable	- Export share
	- Commodity export ratio
	- Financial market size
	- Capital account openness (KAOPEN)
	- Currency pegs (to the dollar and the euro)
	- EU membership dummy
	- Others (commodity export ratio, exchange rate volatility, inflation differential, relative income, etc.)
Key Findings	- The more developed a country's financial markets are, the less likely it is to invoice in US dollars
	- Countries with open capital accounts tend to invoice in euros or their own currency
	- The actual RMB invoicing ratio is less than 10%, but model projections suggest 20–30% is appropriate
# of sample	33–43 countries (up to 50 countries)
	Original dataset combining existing research (Kamps 2006; Goldberg & Tille 2008) with country-level official data
Analysis period	1970–2011 (Euro estimates: 1990–2011)

Summary by the author.

In other prior research, Novy (2006) argued that firms' choice of invoicing currency is influenced by their intention to hedge costs, exchange rate volatility, and currency correlations. Furthermore, with respect to trade involving the United States, he explained that the US dollar is overwhelmingly used as the invoicing currency because raw materials such as oil are priced in US dollars, leading firms to tend to choose the US dollar to hedge their costs. Using a dataset spanning 24 countries and ten currencies, the model was empirically tested, confirming that currency correlations play a significant role in the selection of vehicle currencies such as the US dollar and the euro.

Amiti, Itskhoki, and Konings (2022) analyzed the determinants of currency choice in exports and imports, as well as the impact of these choices on exchange rates, prices, and quantities, using data on currency selection by Belgian firms. Currency choice is an active decision-making process at the firm level and exhibits temporal persistence. Small firms with low import dependence tend to set prices in euros (the producer's currency), and exchange rate fluctuations are fully reflected in their prices. Conversely, large firms with high import dependence tend to set export prices in foreign currencies (particularly the US dollar) and are less susceptible to fluctuations in the euro exchange rate. The choice of import currency influences the selection of export currency; specifically, firms that import in US dollars tend to set export prices in US dollars as well. The impact of exchange rates on prices is determined not only by factors that affect flexible pricing—such as a firm's import dependence and market share—but also by price rigidity stemming from currency choice. For firms that set prices in foreign currencies, exchange rate fluctuations are not fully reflected in prices due to price rigidity. The impact of price rigidity persists for over a year, gradually diminishing over time. Furthermore, exchange rate fluctuations affect not only prices but also export volumes. While the elasticity of volume with respect to price changes is small in the short term, it becomes significant after one year, suggesting that volume adjustments take time.

Brüggen, Georgiadis, and Mehl (2025, ECB) note that renminbi-denominated transactions have risen sharply since 2015, particularly following the COVID-19 pandemic, escalating geopolitical tensions, and the emergence of US–PRC tensions since 2020. The increase in renminbi-denominated invoices is driven by factors such as the Middle East (energy from Saudi Arabia and the UAE), Africa (resources and

investment), and East Asia (regional supply chains). They also estimated how the share of invoice currencies (USD/EUR/RMB) used in imports and exports is influenced by changes in geopolitical distance, trade volume, and bloc formation, showing that as geopolitical distance increases, the use of the US dollar and the euro tends to decrease, while that of the renminbi tends to rise. Furthermore, in terms of trade blocks (trade block share), the study found that as the proportion of trade within a specific block—such as the EU or the PRC—increases, the use of that block's currency (EUR or CNY) tends to rise. On the other hand, the results indicated that invoice currency selection is influenced more by political, institutional, and strategic factors than by price fluctuations, and that the renminbi is expanding primarily in countries geographically close to the PRC within these bloc-based structures.

The following three points are commonly identified in the above prior research as recent characteristics of invoice currency selection. First, in international trade, invoice currencies are highly concentrated in a small number of major currencies, such as the US dollar and the euro, and it has been repeatedly demonstrated that these currencies possess strong dominance and stickiness. Despite the declining shares of the United States and the eurozone in global trade, the shares of the US dollar and the euro as invoice currencies have remained stable or have even increased, indicating strong inertia in invoice currency selection. Firms have an incentive to price in the same currency as their competitors and supply chain practices, and network effects, through global value chains and strategic competitive complementarity, reinforce and entrench currency selection in a self-perpetuating manner. As a result, the US dollar autonomously maintains its status as the primary vehicle currency. Furthermore, dollar-denominated transactions remain central in major commodity trades such as oil; while instances of transactions using the renminbi and other currencies are increasing, they have not yet led to a significant shift in the global share of invoicing currencies.

Second, the expansion of renminbi usage in international transactions since the 2010s has been steady but limited, a trend that is particularly pronounced in Asia and some emerging economies, including in its share of invoicing currency. Nevertheless, the expansion of RMB-invoiced swap agreements and the development of clearing banks have lowered the institutional and transaction cost thresholds for RMB usage through the formation of settlement networks and the provision of liquidity, significantly boosting its use as an invoicing currency. In recent years, amid rising geopolitical tensions and supply chain restructuring, the use of the renminbi for invoicing has expanded as economic ties with the PRC have strengthened in regions such as the Middle East, Africa, and East Asia.

Third, the choice of invoicing currency is determined by both macroeconomic factors and corporate behavior. The choice of currency is an active, sustained decision-making process for firms, and high import dependence and large firm size strengthen the preference for key-currency-invoiced transactions, such as the US dollar. Furthermore, differences in invoicing currencies affect the real economy through exchange rate pass-through and volume adjustments; it has been shown that countries with a high proportion of dollar-denominated transactions exhibit greater sensitivity in import prices and trade volumes to fluctuations in the dollar exchange rate. Recent empirical studies on the Chinese renminbi have found that geopolitical factors, such as geographic and political distance, also influence currency choice. Geographical and political distance from the PRC affects the use of the yuan, while distance from the United States and Europe affects the use of the US dollar and the euro, respectively. It has been confirmed that as the share of intra-block trade increases, the currency representing that block tends to be used more frequently as the invoicing currency.

3. EXAMINATION AND ANALYSIS OF EXPLANATORY VARIABLES RELATED TO INVOICE CURRENCY CHOICE

In this section, building on prior research and the factors promoting the use of local currencies identified by Tomizawa and Shimizu (2026), we will digitize and examine the explanatory variables for promoting renminbi-invoiced transactions.

3.1 Exchange Rates and Their Institutional Factors

In previous studies, exchange rates have been viewed not as factors that directly influence invoice currency choice but rather as factors that exert an indirect influence through the allocation of foreign exchange risk, price adjustment mechanisms, and the institutional environment. Novy (2006) demonstrates that firms select invoice currencies to hedge against cost fluctuations caused by exchange rate movements, arguing that exchange rate volatility and intercurrency correlations—rather than the level of the exchange rate itself—are key factors driving the preference for the US dollar as a vehicle currency. Furthermore, Boz et al. (2020) demonstrated that the choice of invoicing currency significantly influences the pass-through of exchange rate fluctuations to prices and trade volumes, revealing that countries with a higher proportion of dollar-denominated invoices exhibit greater sensitivity in import prices and trade volumes to dollar exchange rate fluctuations. This highlights the importance of the reverse causal relationship, in which exchange rates affect the real economy through currency choice. Recent research has placed greater emphasis on the impact of exchange rate regimes on invoice currency choice rather than on exchange rate fluctuations themselves. Boz et al. (2025) demonstrated that when a destination country is pegged to a specific currency, firms have a stronger incentive to invoice in that currency, noting that strategic complementarity through the exchange rate regime self-reinforces currency choice. Previous research on the renminbi also suggests that while an exchange rate regime anchored to the US dollar inhibits renminbi-denominated invoicing, the development of institutional and financial infrastructure plays a crucial role in expanding the use of nondollar currencies.

In this paper, we include the exchange rate as one of the explanatory variables, using it as the rate of change of a trading partner's currency against the renminbi. In terms of exchange rates, there are several patterns in prior research, such as using logarithms or volatility, or incorporating bilateral exchange rates between the countries in question or exchange rates against the US dollar. For example, Boz et al. (2025) use the logarithm of the exchange rate against the dollar as an explanatory variable to account for the possibility that the share of invoice currencies is influenced by exchange rate changes. In addition to these accounting issues, previous studies have found that exchange rate volatility and inflation rate differences are important determinants of invoice currency choice. Ito and Chinn (2014) include exchange rate volatility as an explanatory variable, because when currency values fluctuate frequently or significantly, transaction costs rise due to an increase in the exchange rate risk premium; consequently, firms and investors tend to avoid using that currency, preferring more stable “hard currencies,” which leads to a tendency to curb the use of the country's own currency for trade invoicing and settlement. On the other hand, Shimizu et al. (2024) conducted an empirical analysis using Japanese customs data on the yen-denominated share in bilateral trade between Japan and its trading partners, and their results indicate that a weaker yen reduces the yen-denominated share,

suggesting that exchange rates influence the choice of invoice currency through various channels.

Since the aforementioned Chinese data indicate that the renminbi-denominated share expanded as the renminbi appreciated, this paper will use the rate of change in the exchange rate between the trading partner's currency and the renminbi for the renminbi-denominated share under consideration. This approach is aimed at verifying whether market expectations of a stronger renminbi may increase the incentive for firms in other countries to use the renminbi. Another reason for using rates of change rather than logarithmic values is that, since this analysis uses the renminbi-denominated share of 70 countries—including emerging economies—from 2015 to 2023 as a sample, we considered it more appropriate to use rates of change to verify the impact of renminbi appreciation and depreciation on the renminbi-denominated share, thereby eliminating the influence of the absolute magnitude of the exchange rate itself.

With respect to exchange rate regimes, this study includes an anchor currency dummy variable. Ilzetzi, Reinhart, and Rogoff (2019) define an anchor or reference currency as a major currency that countries effectively use as a benchmark for exchange rate stability. In other words, an anchor currency is a foreign currency that a country uses as a benchmark (anchor) to stabilize the value of its own currency, and it is the currency at the center of that country's exchange rate policy. Countries that adopt an anchor currency maintain a policy-driven link to that currency to enhance exchange rate stability and the predictability of trade transactions; consequently, firms can mitigate exchange rate risk by using that currency for their trade transactions. In Boz et al. (2025), the share of exports to countries pegged to a given currency is used as a measure of strategic complementarities. The study shows that when a large proportion of a firm's export destinations are pegged to a specific currency (USD/EUR/RMB), the firm has a stronger incentive to invoice in that currency. In this study, following the approach of Boz et al. (2025), we used the revised dataset (2021) from Ilzetzi, Reinhart, and Rogoff (2019) to create a dummy variable indicating whether a country uses the US dollar, the euro, or a basket as its anchor currency. For example, since Malaysia changed its anchor currency from the US dollar to a basket of the US dollar and the euro in January 2017, the US dollar dummy variable is set to "1" for 2016 and "0" for 2017 and subsequent years. Although this dataset reclassifies the exchange rate regimes and anchor currencies for 194 countries and regions since 1946, since data are only available up to 2019, for the 70 countries included in this empirical analysis, we extended the exchange rate anchor classifications from before 2019 after confirming that there were no changes in the classified anchor currencies in the IMF AREAER (2020–2023 editions) for 2020 and beyond.

According to the classification by Ilzetzi, Reinhart, and Rogoff (2019), as of 2021, 103 countries used the US dollar as their anchor currency, 55 countries employed the euro, and 12 countries utilized a currency basket. No country uses the Chinese renminbi as its anchor currency; even in the case of Mongolia, which has a high degree of dependence on the renminbi, the classification designates the US dollar as the anchor currency rather than a peg to the renminbi. A breakdown of anchor currencies by region is shown in Table 5, with the US dollar being the primary anchor in Asia, North America, and Latin America. As for the euro, it is widely adopted not only in Europe but also in Africa.

**Table 5: Anchor Currencies by Country: Classification by Region
(as of December 2019)**

Region	USD	EUR	Basket	Others	Total
East Asia	4	0	1	1	6
Southeast Asia	7	0	2	1	10
South Asia	6	0	0	2	8
Central Asia	5	0	0	0	5
Middle East	11	0	1	2	14
Europe	6	36	2	5	49
Africa	23	19	3	8	53
North and South America	36	0	0	3	39
Oceania	5	0	3	6	14
Total	103	55	12	28	198

Source: Prepared by the author based on Ilzetzki, Reinhart, and Rogoff (2019, QJE).

3.2 Examination of Capital Market Factors

In this paper, we examine several original variables that may influence the choice of the Chinese renminbi as an invoice currency, in addition to macroeconomic variables. The first is capital market factors. We consider using the number of capital controls each country has introduced and maintains as a variable.

Ito and Chinn (2014) analyzed the impact of capital account liberalization (KAOPEN index) on invoicing currency choice and found that countries with open capital accounts tend to invoice in euros or their own currency. The KAOPEN index is a composite index of four dummy variables (0/1) based on the classification in the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER), and the index is constructed using principal component analysis (PCA) of these dummies. According to Chinn and Ito (2006), the four dummy variables used are based on the presence or absence of the following AREAER categories: (1) restrictions on capital transactions; (2) restrictions on the current account; (3) the existence of multiple exchange rates for capital transactions; and (4) mandatory remittance of export proceeds. The value of the first principal component is used as the KAOPEN Index, with higher values indicating greater capital account liberalization.

Data for the KAOPEN index, known as the "Chinn–Ito Index," are publicly available for 181 countries covering the period from 1970 to 2023. For example, a value of 2.28 indicates that restrictions on capital transactions, etc., are loose (or virtually nonexistent) in that country, while a value of –1.94 indicates the strictest level of regulation (Table 6). For example, while countries such as Japan, the United States, the United Kingdom, and Germany show that regulations are virtually nonexistent, many emerging economies such as Brazil and the PRC exhibit negative values, suggesting that significant regulations are in place.

Table 6: KAOPEN Index and Currency/Capital Controls in Major Countries (2023)

Country	KAOPEN	KA_OPEN (Standardization)	Currency-Related Controls	Including Capital Controls
Data range	- 1.94~2.28	0.00~1.00	0~5	0~15
Japan	2.28	1.00	0	3
US	2.28	1.00	0	4
UK	2.28	1.00	0	3
Germany	2.28	1.00	0	5
Brazil	-1.25	0.16	1	10
PRC	-1.25	0.16	4	13
Hong Kong, China	2.28	1.00	0	0
Republic of Korea	2.28	1.00	2	5
Brunei Darussalam	n.a	n.a	0	2
Cambodia	2.28	1.00	0	5
Indonesia	-0.17	0.42	4	13
Lao PDR	-1.25	0.16	4	14
Malaysia	-0.17	0.42	5	15
Myanmar	-1.25	0.16	5	13
Philippines	-0.06	0.45	5	14
Singapore	2.28	1.00	0	1
Thailand	-0.17	0.42	1	9
Viet Nam	-0.17	0.42	3	12

Source: Prepared by the author based on the Chinn-Ito Index and the IMF, AREAER 2023.

As Tomizawa (2023) has shown, many regulations regarding currency and capital transactions remain in place in Asia, particularly in the PRC and Southeast Asia. Similarly, when examining Japan and Western countries based on AREAER data (even though the KAOPEN Index suggests that no regulations exist or that capital account liberalization is high, as mentioned above), it appears that monitoring has been strengthened in some areas in recent years—particularly regarding capital inflows (inward investment) in the fields of direct investment and securities investment—through the establishment of licensing systems and screening mechanisms. When the author counts the number of such regulations (as of 2023) using AREAER, as shown in the right two columns of Table 6, it becomes clear that even in countries considered to have a high degree of capital account freedom, some capital controls remain (for example, Japan has three and the United States has four).

Here, a key point to consider when determining whether a country permits local-currency-denominated current or capital account transactions is which currency and foreign exchange regulations have been introduced for use by both that country and its counterparties. In other words, the critical factor is the extent to which both countries permit transactions and settlements in foreign, or their own, currencies (denominated), including cross-border transactions. In regard to the renminbi, in the context of relations between the PRC and ASEAN countries, issues such as whether renminbi accounts can be opened in individual countries, whether the renminbi can be used in cross-border transactions (settlements), and whether it can be exchanged for local currencies have been summarized in a book (*Renminbi ASEAN* (2021), Table 7).

Table 7: Regulatory Status Regarding the Use and Holding of the Renminbi in ASEAN Countries

Singapore	• There are no strict foreign exchange controls, and no explicit policy restrictions on the use of the RMB. Singapore residents can open RMB accounts both within and outside Singapore. These accounts allow for cross-border RMB remittances and receipts. • Nonresidents of Singapore can open RMB accounts within Singapore. After exchanging Singapore dollars for RMB, they can transfer (move) funds.
Malaysia	• There are no strict foreign exchange controls, and Malaysian residents can open RMB accounts both within and outside Malaysia. These accounts allow for cross-border RMB remittances and receipts. Residents can exchange Malaysian ringgit for RMB. • Nonresidents of Malaysia can also open RMB accounts in Malaysia.
Thailand	• Thai residents can open RMB accounts both within and outside Thailand. These accounts allow for cross-border RMB remittances and receipts. Thai residents can exchange Thai baht for RMB. • Foreign companies can open RMB accounts in Thailand. These accounts allow for the receipt of RMB from abroad.
Indonesia	• Both Indonesian residents and Nonresidents can open RMB accounts in Indonesia. Cross-border RMB settlements can be made using funds in these accounts.
Philippines	• There are no explicit policy restrictions on the use of the RMB, and Philippine residents can open RMB accounts both domestically and abroad. These accounts can be used to send and receive RMB. • Nonresidents of the Philippines can open RMB accounts. These accounts allow for RMB cash withdrawals and cross-border payments.
Viet Nam	• For trade in goods and services in border regions with Vietnam, import settlements using the RMB are permitted. • Enterprises and individuals in border regions with Vietnam may open RMB accounts at banks licensed for foreign exchange operations. Nonresidents of Vietnam are not permitted to open RMB accounts in Vietnam.
Brunei Darussalam	• There are no strict foreign exchange controls, and Brunei residents can open RMB accounts both within and outside Brunei. Cross-border RMB remittances and receipts are possible through these accounts. • Nonresidents of Brunei can also open RMB accounts within Brunei.
Cambodia	• There are no strict foreign exchange controls, and Cambodian residents can open RMB accounts both within and outside Cambodia. These accounts allow for cross-border RMB remittances and receipts. Settlements using RMB are possible between residents, and RMB can be freely exchanged for Cambodian riel. • Nonresidents of Cambodia can also open RMB accounts within Cambodia. RMB can be exchanged for Cambodian riel, and riel exchanged for RMB can be freely taken out of the country.
Lao PDR	• Local currency settlements using the Lao kip and the RMB are already possible for all current and capital account transactions.
Myanmar	• Banks licensed to conduct foreign exchange operations can handle international settlements and payments using the RMB. • Banks licensed to conduct foreign exchange operations can open RMB accounts, but other institutions and individuals cannot open such accounts.

Source: Provisional translation from *Renminbi ASEAN* (2021).

In this study, we first organized information from AREAER data to examine whether any regulations or licensing requirements exist for current account (imports and exports) and capital account transactions in each country. A summary of the situation in the PRC and the ten ASEAN countries is presented in Table 8.

With regard to regulations on current account transactions, requirements such as export or import licenses exist primarily in developing countries. However, as representative items related to the use of other currencies—which are frequently encountered in past practical experience—we included whether there is an obligation to repatriate (remit) export proceeds (foreign currency) to the home country upon export, as well as verification of actual demand restrictions (confirmation of the existence of evidence for remittance) when making import payments.

Table 8: Regulations on Currency and Transactions in the PRC and Selected ASEAN Countries (as of 2023)

		PRC	Cambodia	Indonesia	Malaysia	Philippines	Thailand	Vietnam
Article 8	Countries approved year	1996	2002	1988	1968	1995	1990	2005
Currency	Use of domestic currency for CA transactions			●	●	●		
	Use of domestic currency for capital transactions	●		●	●	●		●
	Use of foreign currency among(including the obligation to use the national currency in domestic transactions)	●		●	●	●		●
	Residents' holding of foreign currency and local currency accounts abroad	●		●	●	●		●
	Non-residents' holding of foreign currency and local currency accounts in Japan	●			●	●	●	
Current Account	Regulations on the repatriation and use of foreign currency proceeds from exports	●		●	●		●	●
Transactions	Evidence of foreign currency payments for imports		●	●	●	●		●
Capital Transactions	Inward Direct Investment	●	●	●	●	●	●	●
	Outward Direct Investment	●			●	●		●
	Inward Securities Investment: Non-residents' Investment in Domestic Stocks and Securities	●	●	●	●	●	●	●
	Outward portfolio investment: Residents' investment in foreign stocks and securities	●		●	●	●	●	●
	Non-residents' trading of domestic derivatives (NDFs, etc.)	●		●	●	●	●	
	Residents' trading of foreign derivatives (NDFs, etc.)	●		●	●	●	●	●
	Commercial and financial loans from non-residents to domestic residents	●	●	●	●	●	●	●
	Commercial and financial loans from domestic residents to non-residents	●	●	●	●	●	●	●

In extracting currency and foreign exchange regulations from the items summarized in the IMF AREAER, among the key issues are:

- (1) whether each country's national currency can be used in current account transactions with foreign countries,
- (2) whether each country's national currency can be used in capital transactions with foreign countries,
- (3) whether foreign currency can be used (for settlement) in domestic transactions (or whether settlement is limited to the domestic currency),
- (4) whether residents can hold foreign currency or domestic currency accounts abroad,
- (5) and whether nonresidents can hold accounts in foreign currency or the domestic currency.

Note: ● indicates that regulations, licensing systems, etc., imposed by the authorities exist for the relevant item.

Source: Prepared by the author based on IMF, AREAER 2023.

With respect to capital transactions, we identified as verification items whether restrictions or licensing requirements exist for each category listed in the individual breakdown of a country's international investment position: direct investment (inbound and outbound), portfolio investment (inbound and outbound), derivatives transactions (inbound and outbound), and other transactions (inbound and outbound commercial loans and loans from financial institutions). As a result, although not shown in the table, while Singapore and Brunei Darussalam have virtually no regulations across the categories (Brunei Darussalam has regulations only for direct investment), countries

such as Malaysia, Indonesia, the Philippines, and Viet Nam² still have regulations in place for many of these categories. On the other hand, Thailand and Cambodia have seen greater deregulation (looser regulations) of currency and capital transactions than other ASEAN countries. Furthermore, as demonstrated by the detailed descriptions of the PRC's regulations in AREAER, even if a country is classified as "Yes" (indicating the existence of some regulation), the degree of deregulation cannot be fully gauged by the simple count of checkmarks. In practice, there are examples where restrictions on investment eligibility for residents and nonresidents in securities investments have been gradually eased, or where regulations have been relaxed in specific areas such as lending and debt guarantees.

The extent to which such regulations exist indicates the degree of freedom with which a country's currency, or the currency of the counterparty, can be used in international transactions. If such transactions are not permitted, it becomes difficult for companies located in that country to conduct trade and capital transactions denominated in renminbi or their own currency. This study counts the five currencies and foreign exchange regulations mentioned above—which are considered most likely to directly influence the choice of invoice currency—and uses them as variables. Specifically, for these five items, we checked the presence or absence of regulations for each country as listed in AREAER, counted them (1 for "regulated," 0 for "unregulated"), and used this as a variable for currency and foreign exchange regulations (minimum 0 to maximum 5, where a score of 5 indicates the strictest currency and foreign exchange regulations).

Table 9: Number of Countries with Remaining Restrictions on Currency and Capital Transactions (as of 2023, 70 Countries Surveyed)

Subject	Items	# of Countries
Currency Transactions	Use of domestic currency for current account transactions	13
	Use of domestic currency for capital transactions	14
	Use of foreign currency (including the obligation to use the national currency in domestic transactions)	20
	Residents' holding of foreign currency and local currency accounts abroad	17
	Nonresidents' holding of foreign currency and local currency accounts in Japan	10
CA transactions	Regulations on the repatriation and use of foreign currency proceeds from exports	17
	Evidence of foreign currency payments for imports	19
Capital Transactions	Inward Direct Investment	58
	Outward Direct Investment	18
	Inward Securities Investment: Nonresidents' Investment in Domestic Stocks and Securities	49
	Outward portfolio investment: Residents' investment in foreign stocks and securities	43
	Nonresidents' trading of domestic derivatives (NDFs, etc.)	25
	Residents' trading of foreign derivatives (NDFs, etc.)	35
	Commercial and financial loans from Nonresidents to domestic residents	19
	Commercial and financial loans from domestic residents to Nonresidents	31
Summary	Average number of regulations per country for currency and exchange controls (5 items)	1.06 points
	Countries with zero currency and exchange controls (5 items) (no controls)	41 countries
	Average number of regulations per country for currency, CA, and capital account (15 items)	5.54 points
	Countries with zero currency, current account, and capital controls (15 items)	4 countries

Source: Prepared by the author based on IMF, AREAER 2023.

² As regards Viet Nam, as shown in Table 8, AREAER classifies the country as having no restrictions on nonresidents holding foreign currency or local currency accounts within Viet Nam since 2015 (the earliest year for which data were available). However, Table 7 indicates that nonresidents in Viet Nam cannot open RMB accounts within the country, indicating a discrepancy.

Summarizing these currency regulations as of 2023 (Table 9), out of 70 countries, 13 have restrictions on the use of their national currency in current account transactions, 14 have restrictions on its use in capital account transactions, 20 have currency restrictions on domestic transactions (settlements), 17 have restrictions on residents holding accounts in foreign or domestic currency within the country, and ten countries have restrictions on nonresidents holding accounts in foreign or domestic currency within the country. Furthermore, as regards the five currency and exchange controls items, the simple average score across 70 countries was 1.06 (41 countries had no restrictions on all five items). However, when the scope was expanded to 15 items—including restrictions on current and capital account transactions—the average rose to 5.54, as the number of restricted items increased (only four countries had no restrictions on all 15 items). In particular, while European countries generally had few regulations, the period since 2015 has seen many countries impose restrictions on domestic investment by nonresidents, including inward direct and inward portfolio investment.

Table 10: Comparison of the KAOPEN Index and the Number of Currency and Exchange Controls in BRICS and ASEAN Countries

Country	Category	2016	2017	2018	2019	2020	2021	2022	2023
PRC	KAOPEN	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25
	# of control	5	5	5	4	4	4	4	4
Brazil	KAOPEN	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25
	# of control	5	5	5	5	5	5	5	5
Russian Federation	KAOPEN	0.83	0.58	0.33	0.08	-0.17	-1.25	-1.25	-1.25
	# of control	1	1	1	1	1	1	3	3
South Africa	KAOPEN	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25
	# of control	5	5	5	5	5	5	5	5
India	KAOPEN	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25	-1.25
	# of control	4	4	4	4	4	4	4	4
Thailand	KAOPEN	-1.25	-1.25	-1.25	-0.17	-0.17	-0.17	-0.17	-0.17
	# of control	2	2	2	2	1	1	1	1
Malaysia	KAOPEN	-0.17	-0.17	-0.17	-0.17	-0.17	-0.17	-0.17	-0.17
	# of control	5	5	5	5	5	5	5	5
Indonesia	KAOPEN	-0.17	-0.17	-0.17	-0.17	-0.17	-0.17	-0.17	-0.17
	# of control	4	4	4	4	4	4	4	4

Source: Prepared by the author based on the Chinn-Ito Index and the IMF, AREAER 2023.

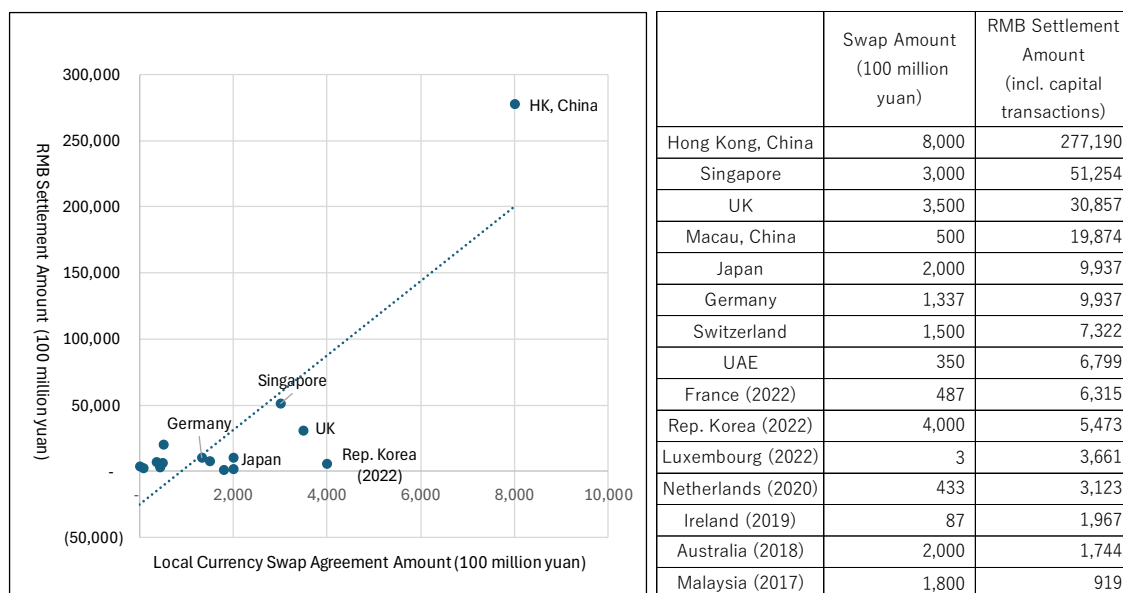
Here, we would like to clarify the difference between the KAOPEN Index and the number of currency and exchange controls used in this study. Table 10 compares the two figures for the BRICS and ASEAN countries. While the KAOPEN Index shows the same regulatory level of -1.25 for all five countries, the figures for currency and exchange controls adopted in this paper reveal differences: Brazil and South Africa are “5,” the PRC and India are “4,” and the Russian Federation is “3.” In regard to the Russian Federation, while regulations have gradually tightened under the KAOPEN Index (from 0.83 in 2016 to -1.25 from 2021 onward), currency and exchange controls have also become stricter, with the value increasing from “1” to “3” since 2022. Furthermore, for Thailand, Malaysia, and Indonesia, the figures are -0.17 , suggesting that capital controls are slightly less stringent than in the BRICS nations. However, for currency and foreign exchange controls, Malaysia is rated “5” and Indonesia “4,” figures that are consistent with their history of strictly limiting the domestic use of both

their own and foreign currencies.³ On the other hand, Thailand has seen progress in deregulation on the KAOPEN Index, achieving the same level of freedom as the other two countries, while currency and exchange controls have also been eased from “2” to “1” since 2020.⁴ Thus, since the count value for currency and exchange controls is expected to indicate a more direct relationship with the use of a country’s currency as an invoice and settlement currency than the KAOPEN Index, we tested it as a substitute variable for KAOPEN in this empirical analysis.

3.3 The PRC’s Policies: Currency Swaps and Overseas Financial Institutions (Financial Infrastructure)

Perez-Saiz and Zhang (2023, IMF) found that the presence of local-currency swap agreements (a dummy variable) is significantly associated with higher renminbi remittance data in SWIFT data. In this paper, we reorganized the data on local-currency swap agreements by the year in which the PRC and each country concluded (or renewed) such agreements, and by the agreed-upon amount (swap ceiling), to examine this relationship more closely.

Figure 1: Relationship Between RMB Swap Agreement Amounts and Renminbi Settlement Volumes (Including Capital Transactions) (2023)



Authors' calculation.

When the swap amounts between the renminbi and the counterpart economy’s currency are listed in RMB terms, and their relationship with the renminbi settlement amount (including capital transactions) with that economy is examined, a consistent relationship can be observed, as shown in the distribution chart in Figure 1: As the local

³ In Indonesia, as discussed in Chapter 2, since 2022, a framework has been established to permit trade contracts and settlements denominated in rupiah and the use of rupiah in international activities, and policies regarding incentive schemes have been outlined.

⁴ The finding that Thailand has the most relaxed regulations among these three countries aligns with the author’s own observations and also corresponds with the fact that the use of the Thai baht has grown the most as part of the progress in the Local Currency Settlement Framework (LCSF) initiatives among these three countries, as discussed in Tomizawa and Shimizu (2026).

currency swap amount increases, so too does the renminbi settlement amount. However, since these settlement amounts include capital transaction figures, and as the data are only published for the top dozen or so economies each year (and only as a share of total RMB receipts and payments), since 2024 onward (RMB Internationalization Report, 2025), even these shares are no longer published. For example, the Republic of Korea recorded a 3.9% share in 2018, ranking fourth behind Hong Kong, China; Singapore; and Germany; but its share fell to 1.3% in 2022, and it was excluded from the published data in 2023 (ranking 12th or lower).

Furthermore, when viewed in terms of the ratio of the swap ceiling amount to the value of imports from the PRC (swap ceiling amount/value of imports from the PRC), as shown in Figure 1, relatively large swap amounts—exceeding 1—have been made available to countries that have faced economic crises in the past, such as Argentina, Iceland, and Mongolia. In the event of a liquidity crisis in each country, this is expected to increase RMB-invoiced imports from the PRC and RMB lending by financial institutions in the partner country.

Table 11: Relationship Between Renminbi Currency Swap Agreement Amounts and Import Values from the PRC

Country	Currency Swap (Size) 1 billion yuan	Currency Swap (Size) 1 million USD	Imports from PRC 1 million USD	Swap/Import Coverage
Argentina	130	18,359	14,463	1.3
Australia	200	28,245	69,531	0.4
Austria	11	1,498	16,336	0.1
Brazil	190	26,833	53,193	0.5
France	52	7,299	78,620	0.1
Germany	134	18,878	169,681	0.1
Hungary	40	5,649	10,338	0.5
Iceland	3.5	494	835	0.6
Indonesia	250	35,306	62,805	0.6
Japan	200	28,245	173,910	0.2
Rep. Korea	400	56,490	143,119	0.4
Macau, China	30	4,345	3,392	1.3
Malaysia	180	25,421	56,806	0.4
Mongolia	15	2,118	3,760	0.6
Russian Federation	50	7,061	43,356	0.2
South Africa	30	4,237	22,019	0.2
Thailand	70	9,886	70,846	0.1
Turkey	35	4,943	44,989	0.1
UK	350	49,429	84,715	0.6

Authors' calculations.

On the other hand, one factor that may have contributed to the poor fit of the variable based on actual swap amounts into the model is the existence of the 350 billion yuan/45 billion euro agreement concluded between the People's Bank of China and the ECB in 2013 (which has been renewed four times for the same amount). The role of this agreement is described as a backstop for supplying RMB liquidity to euro-area banks in the event of turmoil in the renminbi market. While it is necessary to allocate

this swap amount to each country for calculation purposes, we used each euro area country's ratio of bilateral trade volume with the PRC (total imports and exports) to allocate the estimated swap amount. However, these swap agreements have not necessarily been activated, and there may be little correlation between the swap amounts and trade volume with the PRC.

In addition to local-currency swap agreements, other policy variables that encourage the use of the Chinese renminbi in trade and investment, such as clearing banks (dummy) and CFETS direct exchange amounts, can be considered candidates. However, the direct exchange volume is overwhelmingly dominated by RMB–USD transactions. Furthermore, among the currencies with relatively high trading volumes, the dataset includes the euro (as mentioned above), as well as the Canadian dollar and the Singapore dollar—countries for which the renminbi's share is not reported due to limitations in the Boz et al. dataset. For these reasons, we judged that these variables have low validity as variables for this empirical study. Furthermore, although it is presumed that the number of financial institutions connected to CIPS via indirect links is strongly correlated with the volume of RMB settlements, the published data of CIPS are limited to aggregate figures by continent, making them difficult to use as a variable in this analysis.

**Table 12: Policy Support for Promoting RMB Transactions and Settlements
(End of 2025)**

Local Currency Swap Agreements	42 economies and regions (currencies)
	Hong Kong, China: 800 billion yuan (940 billion HKD)
	ECB 350 billion yuan (45 billion euros)
	Japan: 200 billion yuan (3.4 trillion yen)
Renminbi Clearing Banks	31 economies and regions, 35 banks
	2 in the US, 2 in Japan, 3 in the EU
Shanghai CFETS Direct Exchange	25 economies and regions (currencies)
	In addition: US dollar, Hong Kong dollar
	Regional trading currencies: 4 (+1)
CIPS Cross-Border RMB Settlement System	124 economies and regions
	Direct participants: 193
	Indirect participants: 1,573

Source: Authors' counting from each related website.

Therefore, this study focuses on capturing and counting the overseas expansion and activities of Chinese-owned financial institutions. This expansion is considered to support RMB business and RMB financial infrastructure in the host countries. As an example, as cited in the CEPR (2025) article, in 2023, Argentina faced a severe dollar shortage, and as its foreign exchange reserves were depleted, there was a rapid expansion of RMB-invoiced settlements. The share of such settlements rose from nearly 0% in early 2023 to approximately 50% by September of that year, covering a wide range of sectors, including electronics, auto parts, apparel, and oil and mining companies. This development was driven by the significant expansion of the renminbi–Argentine peso currency swap line with the People's Bank of China in early 2023, a portion of which was activated.

It has also been noted that, amid a severe dollar shortage, companies found it difficult to procure dollars, making RMB settlements a practical alternative; furthermore, companies that had been transacting with banks with high ratios of USD-denominated

loans were slower to switch to the renminbi. Partly due to strong government support for renminbi settlement, companies that already had business relationships with the Industrial and Commercial Bank of China (ICBC)⁵ reportedly had a significantly higher proportion of renminbi-denominated transactions than other companies. A ripple effect was also observed, in which larger companies were more likely to use the renminbi, and firms that began using it for transactions with the PRC tended to use it for imports from other countries as well.

On the other hand, following the change in government in November 2023, the use of the renminbi plummeted amid improved dollar supply and changes in exchange rate policy, falling to less than 10% as of March 2024. This example demonstrates that if the institutional foundation of Chinese banks' overseas networks is established and bilateral relations progress smoothly, the internationalization of the renminbi can proceed in accordance with surrounding conditions. Behind this development, factors such as deregulation allowing local residents (domestic corporations) and others to use the renminbi as a trading currency, the formation of offshore renminbi and interbank markets centered on clearing banks capable of procuring renminbi locally, and the existence of local currency swap agreements between the renminbi and the partner country's currency—which underpin these markets—are also crucial.⁶

Table 13: Argentina's Relationship with the PRC

Local Currency Swap Agreements	Signed in 2009: 70 billion yuan / 38 billion Argentine pesos Renewed in 2014: 70 billion yuan / 90 billion Argentine pesos Renewed in 2017: 70 billion yuan / 175 billion Argentine pesos Renewed in 2020: 70 billion yuan / 730 billion Argentine pesos Additional 60 billion yuan Renewed in 2023: 130 billion yuan / 4.5 trillion Argentine pesos Of which 35 billion yuan has been activated
Renminbi Clearing Banks	2015: Establishment of Industrial and Commercial Bank of China (ICBC Argentina)
Shanghai CFETS Direct Exchange	n.a.
Entry of Chinese Financial Institutions	2012: ICBC acquired 80% of Standard Bank's shares 2013: ICBC established a local subsidiary in Argentina 2019: Bank of China opened a branch in Buenos Aires
High-level Exchanges	2010: President Kirchner and President Hu Jintao (Beijing) 2012: President Kirchner and Premier Wen Jiabao (Buenos Aires) 2014: President Kirchner and President Xi Jinping (Buenos Aires) 2017: President Macri and President Xi Jinping (Beijing) 2022: President Fernández and President Xi Jinping (Beijing)
MOU on E-Commerce Cooperation	Signed in 2018

Source: Authors' counting from each related website.

⁵ The Industrial and Commercial Bank of China (ICBC) is the sixth-largest lender in Argentina, with over 100 branches in the country (CEPR 2025). <https://nic.icbc/ICBC/EN/PersonalFinance/CrossborderFinancialServices/OverseasFinancialService/ICBCArgentina/> (ICBC website).

⁶ The CFETS in the PRC does not offer direct exchange for the Argentine peso.

As shown in Table 13, Argentina first entered into a currency swap agreement with the PRC in 2009 (worth 70 billion yuan), which was supplemented by an additional 60 billion yuan in 2020, and a swap agreement worth 130 billion yuan (4.5 trillion pesos) has been in place since 2023. Out of the total swap amount, 35 billion yuan was activated in 2023.⁷ The Industrial and Commercial Bank of China (ICBC) established a local subsidiary in Argentina in 2013 and was designated as an RMB clearing bank in 2015. Aside from the ICBC, the Bank of China entered the market in 2019 as a branch (a branch of its head office in the PRC).⁸

Commercial banks' entry into various countries takes the form of local subsidiaries, branches of their Chinese headquarters, or representative offices. A key characteristic is that local subsidiaries offer the greatest flexibility in branch operations, whereas branches may face restrictions on the scope of business activities permitted by authorities (for example, they may be allowed only to extend foreign-currency loans in the host country, but not local-currency loans or retail deposit services). Furthermore, in many cases where only a representative office is established, local business operations are not permitted, and only activities such as information gathering and business matching are allowed.

Taking these practical advantages and constraints into account, we scored Chinese banks entering various countries as “3” if they had a local subsidiary by the end of that year, “2” if they had a branch of their Chinese head office, and “1” if they had a representative office. In the case of Argentina mentioned above, since the Industrial and Commercial Bank of China (ICBC) entered the market in 2013, the period from 2015 to 2018 was scored as “3”; and since a Bank of China branch was established in 2019, “2” was added, resulting in a total count of “5” for the period from 2019 to 2023.

To verify market entry, we used Moody's Bank Focus database to extract data on the so-called “Big Five” Chinese banks—the Bank of China, the Industrial and Commercial Bank of China, the China Construction Bank, the Agricultural Bank of China, and the Bank of Communications—and cross-referenced this with information on their websites regarding their operational structures. We further supplemented the data by identifying the year in which each financial institution's local office was established, based on publicly available information.

Table 14: Top 5 Chinese Banks' Overseas Offices and Shares

Area	2015		2023	
	Number of Locations	Market Share	Number of Locations	Market Share
Asia	133	45.9%	166	43.8%
Europe	96	33.1%	123	32.5%
Africa	15	5.2%	25	6.6%
North America	20	6.9%	20	5.5%
Latin America	26	9.0%	44	11.6%
Oceania	20	6.9%	21	5.5%
Total	290		379	

Source: Compiled by Tomizawa based on Bank Focus and each bank's website.

⁷ <https://buenosairesherald.com/economics/argentina-renews-us5-billion-currency-swap-with-china-for-12-months>.

⁸ Benguria and Novy (2025) reported the recent usage of renminbi in Argentina.

Summarizing the results, as shown in Table 14, we found that expansion into the Asian region accounts for about half of the total, followed by expansion into Europe, while expansion into Africa is limited. Until around 2010, overseas branches of Chinese banks were primarily established by the Bank of China and the Industrial and Commercial Bank of China in so-called “financially advanced countries” (regions), but since the 2010s, the other three banks have also expanded their branch networks. In step with the renminbi’s internationalization, expansion into countries along the Belt and Road route has increased, particularly since 2015. It is presumed that this is driving renminbi-denominated transactions in trade and investment through trade finance and other corporate renminbi lending activities.⁹ While the pace of expansion and branch openings accelerated from 2015 to 2019, the rate of new openings has slowed following the COVID-19 pandemic. With respect to the balance of market share across regions, although the absolute number of bases in Asia increased over the eight years up to 2023 (+33), the market shares of Latin America and Africa have grown.

3.4 Other PRC Government-Led Policies: Number of Visits by PRC Government Officials and Their Counterparts, and Other Related Policies

In prior research, Boz et al. (2025, IMF) and Perez-Saiz and Zhang (2023, IMF) used voting behavior at the UN General Assembly as an indicator of political proximity to the currency issuer. They demonstrated that a greater political distance (i.e., lower political proximity) is negatively correlated with the use of that country’s currency. In this paper, rather than using variables indicating geopolitical distance or relations with the PRC, we used PRC-led policies aimed at strengthening relations with the PRC as explanatory variables to promote the use of the renminbi.

(1) Number of visits by PRC government officials and their counterparts

In this paper, as candidate variables indicating such geopolitical distance or relationships, we first considered whether a country is a member of Global South groupings such as BRICS and the SCO. In fact, within these groups, measures such as the use of local currencies in transactions among member states and the development of independent settlement infrastructures are being considered to promote transactions denominated in local (domestic) currencies; thus, a certain correlation with the choice of invoicing currency is assumed. However, these groups are still in the developmental stage, have few participating countries, and show limited overlap with the countries in this panel dataset, raising concerns over an insufficient sample size.

Another variable to consider is whether a country participates in the PRC’s Belt and Road Initiative. To increase RMB-invoiced transactions, the PRC government facilitates foreign exchange trading between the renminbi and 17 “Belt and Road currencies” on platforms such as the CFETS direct exchange market (Report on the Internationalization of the Renminbi 2024); however, the specific currencies are not identified in the report. The report also notes that, as of the end of 2023, 68 financial institutions from Belt and Road participating countries had entered the PRC’s interbank foreign exchange market.

⁹ Although this paper did not cover it in detail, deregulation of corporate cash management through “cash pools” is underway in 16 regions within the PRC, including Beijing and Shanghai (to be expanded in March 2025). Adoption is progressing primarily among multinational corporations, and since 2021, initiatives for cash pools that integrate both foreign currencies and the renminbi have been promoted, with Chinese financial institutions providing active support.

According to the above report (2023), the value of cross-border renminbi payments between the PRC and countries jointly building the Belt and Road Initiative reached 7.1 trillion yuan (up 4.4% year-on-year), accounting for 16.9% of the total value of cross-border renminbi payments between the PRC and the rest of the world. Of this total, payments related to goods trade amounted to 2.3 trillion yuan (up 75.7% year-on-year), those related to services trade amounted to 230.75 billion yuan (up 35.4% year-on-year), and payments related to direct investment amounted to 780.793 billion yuan (up 19% year-on-year), all showing strong growth. Furthermore, as of September 2023, local currency swap agreements have been concluded with 30 countries jointly building the “Belt and Road” (with agreements in place with 40 countries and regions worldwide at that time), and RMB clearing banks have been established in 17 countries participating in the joint construction of the “Belt and Road” (with 31 banks designated across 29 countries and regions worldwide at that time).

Furthermore, on the “Belt and Road Network” website,¹⁰ which the PRC government uses to promote Belt and Road construction, the number of countries for which Belt and Road-related activities and investment information are provided stands at 156¹¹ (as of 18 February 2026). However, this figure includes countries that have already announced their intention to withdraw from the Belt and Road Initiative (such as Italy in 2023 and Panama in 2025). According to data provided by the Green Finance & Development Center at Fudan University’s Fanhai International School of Finance (FISF), between 146 and 150 countries have signed MOUs with the PRC regarding Belt and Road cooperation, and a specific list has been published; however, the report also notes that there is uncertainty regarding this list. On the other hand, the China Foreign Investment Statistical Bulletin (2018) compiles figures on foreign direct investment (FDI) into the PRC from 62 “Belt and Road countries” as part of a trend analysis of FDI from these nations. However, since some other information sites suggest that the number of Belt and Road countries is approximately 130, it is difficult to determine the definition of participating countries or the accuracy of the figures.

Taking this situation into account, we use the number of leadership-level exchanges between the PRC and each country as a variable representing political proximity, rather than focusing on membership or participation in multilateral organizations. Specifically, as part of the internationalization of the renminbi, we counted the number of overseas visits made by the President of the PRC or the Premier of the State Council since 2010, when renminbi settlements effectively began (note that for President Xi Jinping, visits made after 2010 while he was Vice President [March 2008 – March 2013] were also included in the counts for visits by the President and the Premier). Conversely, for visits to the PRC by leaders from partner countries, we similarly limited the count to exchanges at the presidential or prime ministerial level (extracting visit data from various website reports as of the end of 2025).¹² For example, if the president of Country A visited the PRC in 2010, 2015, and 2017, the score would be “2” for 2016 and “3” for 2017 and beyond. In this empirical model, the total number of visits between the PRC and the partner country was used as a variable. Visits made to attend international conferences were also included in the visit count, as bilateral meetings often take place on the sidelines of such events.

¹⁰ <https://www.yidaiyilu.gov.cn/>.

¹¹ The breakdown is as follows: 41 countries and regions in Asia, 52 in Africa, 27 in Europe, 13 in North America, 11 in South America, and 12 in Oceania.

¹² As regards multilateral conferences and other events held in the PRC, it is difficult to confirm the visits of all foreign leaders to the PRC based solely on news reports; verifying more accurate information through the PRC government documents and other sources will be a task for future research.

In actual diplomacy, foreign ministers generally lay the groundwork prior to heads-of-state visits, followed by summit-level diplomacy. Seizing opportunities for exchanges between heads of state, efforts are made to elevate the level of bilateral partnership and produce outcome documents regarding various forms of cooperation, such as the signing of memorandums of understanding on major infrastructure projects or (in the case of the PRC) the confirmation of cooperation under the Belt and Road Initiative. In the financial sector as well, often a package of measures is advanced in conjunction with, or around the timing of, summit-level exchanges, including the conclusion (new or renewed) of local swap agreements between the renminbi and the partner country's currency, the establishment of renminbi clearing banks, the launch of direct trading on the Shanghai CFETS, and the entry of Chinese commercial banks into the partner country. Such government-driven incentives can be viewed as a “jumpstart,” as discussed in Tomizawa (2023).

For example, around the time of Brazilian President Lula's visit to the PRC in 2023, arrangements were made to enable RMB settlement between the two countries' central banks, and the Brazilian subsidiary of the Industrial and Commercial Bank of China (ICBC) was designated as an RMB clearing bank. In Japan as well, following the 2011 summit, efforts to promote financial cooperation between Japan and the PRC were advanced under the agreement reached there, leading to the introduction of direct yen–RMB trading in both markets. During the next round of exchanges in 2018, a series of developments followed, including the establishment of a renminbi clearing bank in Tokyo, the conclusion (or resumption) of a yen–renminbi currency swap agreement, the approval of a 200 billion yuan investment quota for qualified institutional investors in renminbi (to promote securities investment), and the issuance of “Panda bonds”—renminbi-denominated bonds for nonresidents—by two Japanese megabanks within the PRC. In this way, high-level exchanges not only yield immediate effects—such as strengthening trade and investment cooperation and advancing financial cooperation mechanisms and infrastructure development in the financial and monetary sectors—but also yield long-term benefits. Therefore, I believe it is more natural to count these exchanges using a method that accounts for the increasing number of variables with each visit.

(2) China–Europe Railway Express Dummy

Previous research by Perez-Saiz and Zhang (2023, IMF) uses the geographical proximity (distance) between a capital city and Beijing as an explanatory variable and finds a significant negative relationship with the use of the renminbi in SWIFT settlements (i.e., the greater the distance, the lower the proportion of renminbi settlements). On the other hand, since geographical distance is thought to reflect various factors, such as ease of trade, logistics, financial networks, and cultural proximity, it can also serve as a proxy for these heterogeneous factors.

Therefore, in this study, instead of using numerical values for the geographical distance between capitals, we considered the existence of the “China Railway Express (CR Express),” a freight rail network connecting the PRC with Central Asia and European countries, as a factor. In the PRC, there had long been plans for a railway connecting Lianyungang in Jiangsu Province with Amsterdam in the Netherlands, but in 2011, a route connecting Chongqing to Duisburg, Germany, via Kazakhstan and the Russian Federation, was launched. This freight route has developed into a transportation corridor linking cities in the central and western PRC with Europe, replacing costly air freight.

When President Xi Jinping announced the “Belt and Road” initiative in 2013, this freight railway became a core project of the national strategy, and efforts to expand its routes have been carried out subsequently. At the 2017 Belt and Road International Cooperation Summit, the railway sectors of seven countries—including the PRC, Germany, the Russian Federation, and Poland—formally signed a cooperation agreement to standardize operating rules. Furthermore, in 2019, the International Freight Consignment Note (a single consignment note) was introduced, eliminating the need to recreate documents in each country and significantly enhancing convenience. Even during the COVID-19 pandemic, while other logistics routes stagnated, rail freight transport proved to be well suited to transporting medical supplies and e-commerce goods, leading to an increase in the number of trains and the volume of cargo transported.

Table 15 shows that the China Railway Express currently operates in 128 cities across the PRC, and the service connects 232 cities across 26 European nations and over 100 cities in 11 Asian countries. To date, a cumulative total of 110,000 railcars have been deployed, transporting a cumulative cargo volume of 11.7 million TEUs (20-foot equivalent units) (China–Europe Railway Express Development Report 2025). While it is difficult to pinpoint exactly when the routes began, the service functions as a major logistics artery connecting European countries, Central Asia, and the PRC, with goods being transported (imported and exported) via sea routes from both ends of the network.

Table 15: Countries Connected by the China–Europe Railway Express

European side	Asian side	Note
Germany	PRC	
France	Kazakhstan	
Italy	Uzbekistan	
Spain	Kyrgyz Republic	
Netherlands	Turkmenistan	
Belgium	Iran	
UK	Türkiye	
Poland	Georgia	
Czech Republic	Mongolia	
Austria	Azerbaijan	
Hungary	Viet Nam*	*Countries on the connecting route
Russian Federation	Lao PDR*	*Countries on the connecting route
Slovakia	Thailand*	*Countries on the connecting route
Belarus	Pakistan*	*Countries on the connecting route
Ukraine		
Romania		
Bulgaria		
Serbia		
Sweden**		**Countries of destination for maritime cargo
Finland**		**Countries of destination for maritime cargo
Denmark**		**Countries of destination for maritime cargo
Switzerland**		**Countries of destination for maritime cargo
Norway**		**Countries of destination for maritime cargo

Source: CHINA RAILWAY Express (<https://www.crexpress.cn/zoblmenu/en/allTraffic>).

On the PRC side, the three main routes (the PRC's Three Major Corridors) are the Western Corridor (Alashankou–Korgos), the Central Corridor (via Erenhot), and the Eastern Corridor (Manzhouli–Suifenhe). On the European side, the network consists of the Northern Route (via the Russian Federation), the Central Route (via Belarus and Poland), and the Southern Route (via Türkiye).

Furthermore, as mentioned in Chapter 2 and in Tomizawa (2023), the digital yuan has been used for PRC–Europe freight train operations on the PRC side. While the extent to which these initiatives have progressed is unclear, this freight route plays a crucial role as a key transport infrastructure connecting the Belt and Road Initiative, particularly the Silk Road Economic Belt.

For this reason, to account for geographical distance, a dummy variable was created in which countries through which the China–Europe Railway Express route passes (or is connected) are assigned a value of “1.” Specifically, while the China–Europe Railway Express is said to pass through “26 European countries and 11 Asian countries,” as mentioned above, a detailed list of the countries has not been published. Therefore, we created a list based on data regarding the departure and arrival cities of trains published on the “Belt and Road Network” website, and supplemented it with information from articles on the China–Europe Railway Express published on that website. As a result, on the European side, we designated 23 countries—including five destination countries (such as the Nordic countries) for cargo transported by sea from ports at the end of the railway lines—as target countries for the China–Europe Railway Express dummy variable. On the Asian side, we designated 13 countries—including four countries (such as Thailand and Viet Nam) with routes connecting to the China–Europe Railway Express—as target countries.

(3) Memorandum of Understanding (MOU) on E-Commerce

As explained in Section 2 and in Tomizawa (2023), the PRC and Southeast Asia, in particular, are regions where digital adoption is spreading rapidly, and human mobility is more active than in other regions. We have long believed that data on these factors provide key insights into promoting trade transactions denominated in local currencies.

In regard to digital payments, we believe that the increase in overseas travel by Chinese citizens and business travelers may lead to Chinese mobile payment tools such as Alipay and WeChat Pay playing an expanding role in connecting with overseas markets and facilitating renminbi settlements. Meanwhile, in Southeast Asian countries, efforts are underway as part of the ASEAN initiative to expand networks for instant bilateral settlements using cross-border QR code payments. As for Central Bank Digital Currencies (CBDCs), countries such as the PRC, Thailand, and the UAE are continuing to conduct cross-border connectivity experiments. While it is difficult to predict the pace at which these wholesale digital currency payment initiatives will progress, if a certain level of connectivity is achieved, it is expected to have a significant impact on the existing payment network centered on SWIFT.

As indicated by the ADB (2025), one of the defining characteristics of the Asian region is the high volume of human mobility. Consequently, the diverse development and use of mobile payment methods driven by this factor, as well as their optimization, are expected to advance further. Looking at trends in the number of travelers between the PRC and ASEAN, the number of visitors from the PRC had been growing significantly, particularly before the COVID-19 pandemic. While this trend aligns with rising incomes among Chinese citizens and the expansion of Chinese companies into foreign markets, data on the number of outbound travelers from the PRC are not published on a country-by-country basis (except when released by destination countries such as

ASEAN nations). Furthermore, the PRC's National Bureau of Statistics has not published data on inbound travelers to the PRC from major countries since 2018, making it difficult to quantitatively assess the actual volume of travel to and from the PRC.

While the movement of people slowed amid the COVID-19 pandemic, so-called “e-commerce” saw significant growth in both domestic transactions within the PRC and in trade between the PRC and other countries. According to the China Digital Trade Development Report (2025), Chinese customs statistics show that the scale of the PRC's cross-border e-commerce imports and exports reached 2.71 trillion yuan in 2024, a 14% increase from the previous year, accounting for 6.2% of the total value of goods imports and exports during the same period (up 0.5 percentage points year-on-year). Of this total, exports amounted to 2.15 trillion yuan (up 16.9% year-on-year), with e-commerce accounting for 8.5% of total exports, while imports amounted to 0.56 trillion yuan (up 4.1% year-on-year), accounting for 3% of total imports (Table 16).

Table 16: Share of the PRC's Cross-Border E-Commerce Transactions in Total Imports and Exports

2024	Total trade		E-commerce Transactions		Share of E-commerce transactions in total trade
	Amount (trillion yuan)	Year-on-year Growth Rate	Amount (trillion yuan)	Year-on-year Growth Rate	
Exports	25.45	7.1%	2.15	16.9	8.5%
Imports	18.39	2.3%	0.56	4.1	3.0%
Total	43.85	5.0%	2.71	14.0	6.2%

Source: General Administration of Customs.

With respect to the concept of cross-border e-commerce, the development report defines it as trade involving digital ordering within digital trade,¹³ specifically transactions concluded on online platforms.

Table 17: The PRC's Cross-Border E-Commerce (Import/Export Value and Growth Rate)

	Amount (100 million yuan)			Year-on-Year Growth Rate (%)		
	Imports and Exports	Exports	Imports	Imports and Exports	Exports	Imports
2018	10,557	6,116	4,441	50.0	67.0	31.1
2019	12,903	7,981	4,922	22.2	30.5	10.8
2020	16,220	10,850	5,370	25.7	39.2	9.1
2021	19,237	13,918	5,319	18.6	28.3	-0.9
2022	20,599	15,321	5,278	7.1	10.1	-0.8
2023	23,744	18,409	5,335	15.3	20.2	1.1
2024	27,072	21,520	5,552	14.0	16.9	4.1

Source: General Administration of Customs of China.

¹³ Digital trade is defined as a form of international trade that treats “data” as a core production factor, centers on “digital services,” and is conducted through “digitally ordered” and “digitally delivered” transactions (electronic delivery of services such as software, cloud computing, AI, and consulting) (China Digital Trade Development Report 2025). E-commerce falls under this category of digitally ordered trade and is considered a type of digital trade.

By country of trade partner for e-commerce (2024), exports are dominated by three countries—the United States (36.2%), the United Kingdom (11.7%), and Germany (5.7%)—which together account for 53.6% of the total. In the Southeast Asian market, Viet Nam's share has grown for two consecutive years, reaching 3.7%. In Latin America, Brazil (1.6%) and Mexico (1.5%) account for the largest shares. Imports are led by the United States (15.8%), Japan (10.5%), Germany (9.8%), and Australia (9.6%).¹⁴

The People's Bank of China has indicated that renminbi-denominated e-commerce transactions are also growing rapidly. Although the data are fragmentary, the value of cross-border e-commerce transactions settled in renminbi exhibited significant growth, reaching 258.4 billion yuan in 2020 (a 50% increase year-on-year) and 431.7 billion yuan from January to June 2022 (a 20.7% increase year-on-year) (Report on the Internationalization of the RMB 2021, 2022). According to the authors' calculations, 15.9% of the previously mentioned cross-border e-commerce trade volume in 2020 and approximately 40% in the first half of 2022 were denominated in renminbi.

JETRO (2025) notes that, while payment methods on Chinese e-commerce sites vary by e-commerce model, smartphone-based online payment services such as "Alipay" and "WeChat Pay" are widely used in the PRC. Furthermore, for cross-border e-commerce using dedicated platforms, the payment flow is described as consumer → dedicated platform → seller. The actual details of payment settlements between platforms and sellers—including the currency denomination and settlement currency—are not always clear. On the other hand, Peng (2025) indicates that the proportion of RMB settlements in cross-border e-commerce is not necessarily higher than that in other general trade. Nevertheless, it cannot be denied that the reality of such e-commerce transactions in recent years has bolstered trade in small-value, low-volume consumer goods—particularly those centered on individual consumers—and has served as a driving force for the PRC's overall trade, which is seeing its rapid expansion trend begin to slow. It is natural to assume that renminbi settlements are used to some extent in cross-border e-commerce as well.

Since 2016, the PRC has signed e-commerce cooperation memorandums of understanding (MOUs) with 36 countries to deepen economic and trade cooperation under the Belt and Road Initiative, with a focus on promoting B2B e-commerce and small-scale B2C online transactions. These initiatives aim to facilitate customs cooperation and the revision of other legal systems when conducting online or other electronic transactions with partner countries, and to foster industrial cooperation (business matching).

Table 18 summarizes the countries with which the PRC has signed Memoranda of Understanding (MOUs) on e-commerce as of February 2026, along with the year each agreement was signed. The PRC's objectives in signing e-commerce MOUs with these countries include jointly building global e-commerce markets, facilitating online connections between companies from both sides through business matching, and creating an environment that facilitates cross-border e-commerce trade. Additionally, the goal is to achieve deregulation and institutional coordination (institutional opening) in order to facilitate market entry. By standardizing systems such as electronic bills of lading (e-BL) and electronic invoices, establishing platforms, and digitizing cross-border

¹⁴ By product category, consumer goods account for 97.5% of exports, comprising clothing, footwear, and jewelry (22.6%), household and office electrical appliances (15%), and digital devices (13.9%). Consumer goods account for 97.8% of imports, including beauty cosmetics, perfumes, and detergents (27.1%), fresh food (23.5%), pharmaceuticals, health products, and medical devices (15.4%), and powdered milk (10.8%).

procedures to reduce procedural costs, the aim is to accelerate cross-border e-commerce. Furthermore, to enhance the digital capabilities of partner countries, the PRC is providing e-commerce technical assistance and human resource development, particularly in Africa. Furthermore, as part of interregional cooperation, PRC local governments are collaborating with partner countries to establish “e-commerce production and supply chains” (for agricultural and industrial products) and to coordinate logistics efforts. These activities are driving the growth of cross-border online retail and e-commerce, while also advancing logistics infrastructure—including the China–Europe Railway Express—and the development of digital customs clearance systems.

Table 18: Countries with which the PRC Has Signed Memorandums of Understanding (MOUs) on E-commerce (As of February 2026, 36 Countries in Total)

Europe (8)		Central Asia (3)		Southeast Asia (7)		Latin America (5)	
Serbia	2024	Tajikistan	2024	Indonesia	2023	Colombia	2019
Belarus	2022	Uzbekistan	2019	Philippines	2023	Panama	2018
Italy	2019	Kazakhstan	2018	Lao PDR	2022	Argentina	2018
Iceland	2018	South Asia (2)		Thailand	2022	Brazil	2017
Russian Federation	2018	Bangladesh	2025	Singapore	2022	Chile	2016
Austria	2018	Pakistan	2022	Cambodia	2017		
Hungary	2017	Middle East (3)		Viet Nam	2017		
Estonia	2017	Bahrain	2023 ?	Oceania (4)			
Africa (4)		United Arab Emira	2018	Vanuatu	2019		
Egypt	2025 ?	Kuwait	2018	Samoa	2019		
Kenya	2025			Australia	2017		
Senegal	2021			New Zealand	2017		
Ruwanda	2018						

Source: Compiled from the National E-Commerce Public Service Network of the Ministry of Commerce of China.

As online retail (cross-border e-commerce), market sales via platforms (B2C), and parcel and express logistics from the PRC to overseas (EMS, cross-border express) have increased, the total value of cross-border e-commerce imports and exports in 2024 was projected to reach 2.7 trillion yuan, as previously mentioned. This represents a 14% year-on-year increase, surpassing the PRC’s overall imports and exports growth rate of 5%. Interindustry collaboration has also led to growth in e-commerce beyond online retail (B2B trade). For example, for the ordering and procurement of agricultural products, 65 production bases across 19 countries have been established as e-commerce supply sources through partnerships with overseas agricultural production bases. Furthermore, the digitization of trade—including logistics and customs clearance—is advancing, creating a framework for the expansion of total transaction volume.

While the full breakdown of the PRC’s digital trade and e-commerce volumes by country remains unclear, it is reasonable to assume that the signing of such MOUs, backed by the PRC’s production and consumption capacity, is somehow linked to the proportion of RMB-invoiced transactions in trade and investment. For example, under these initiatives, efforts are underway with Thailand to link the online sale of Thai durians and other products to the Chinese market, enabling rapid customs clearance. It is believed that Chinese consumers generally pay in renminbi for purchases made on

online platforms, and as the volume of transactions with partner countries increases, this could lead to a trend toward promoting transactions in the local currencies of both countries or direct currency exchange. While previous studies have rarely used digital variables in the choice of local currency for invoicing, this study adopted a dummy variable in which countries that have signed e-commerce MOUs are assigned a value of “1.”

4. EMPIRICAL ANALYSIS OF FACTORS PROMOTING RMB-INVISED TRANSACTIONS

As explained in the literature review, empirical analyses of the determinants of trade invoicing currency selection have been conducted using various datasets. This paper attempts an empirical analysis of the factors that have promoted RMB-invoiced transactions in each country.

4.1 Examination of Dependent Variables

In each country, the share of trade with the PRC (exports and imports) in total global trade is undeniably on an overall upward trend. This can be attributed to the PRC’s growing trade volume, particularly the increasing role of the PRC as a final consumption destination driven by rising incomes and purchasing power among Chinese citizens (individuals), as well as improvements in the manufacturing capabilities of Chinese enterprises. On the other hand, as pointed out in some of the prior studies cited in the previous section, it is natural to assume that the PRC’s external trade relations are not solely characterized by an increase in market share but also include factors (headwinds) that reduce the PRC’s share, against the backdrop of geopolitical trends—including changes in US–PRC relations—a shift in supply chains, and the overseas expansion of Chinese enterprises.

When viewed from the perspective of invoicing currency, the share of renminbi-invoiced transactions in each country’s total exports and imports generally falls within the range of each country’s share of bilateral trade with the PRC, as confirmed by the country-specific data in Tomizawa and Shimizu (2026). Thus, for all countries, the use of the renminbi appears to be limited, for the time being, primarily to bilateral trade with the PRC.

Based on this approach, for example, Japan’s imports from the PRC totaling \$173,910,000 account for approximately 22.1% of its total imports of \$787,198,000 (2023). In contrast, the share of the renminbi in Japan’s total trade, measured by invoice currency, is estimated at 1.70% by Boz et al. (2025). This amounts to approximately \$13,382,000 ($= 787,198 \times 0.0170$), meaning that, in terms of bilateral imports, about 7.7% of imports from the PRC ($= 13,382 / 173,910$) are invoiced in renminbi. In contrast, the currency breakdown of trade transactions published by the Japanese Ministry of Finance shows that for bilateral trade with the PRC in the first half of 2023, the share of imports from the PRC denominated in renminbi was 7.6%, and 7.4% in the second half. Furthermore, for imports from Hong Kong, China, the share was 0.1% in the first half and 0.2% in the second half. Combined, this totals 7.7% for the first half and 7.6% in the second half, which generally aligns with the estimates mentioned above.

On the other hand, as regards the use of the renminbi in import transactions in Brazil, it is a fact that the renminbi is beginning to be used not only in bilateral trade between the PRC and Brazil but also in transactions between Brazilian companies and firms in economies such as Japan; Taipei, China; Southeast Asia; and the Russian Federation. However, given that the use of the renminbi in global transactions remains limited at this point, it is currently more natural to assume that, in most cases where the renminbi is used as an invoice currency, it is selected and utilized within the context of that economy's (or company's) bilateral trade with the PRC.

As part of the empirical analysis in this paper, we first considered using the share of invoice currencies (including those denominated in the renminbi) for each economy, based on publicly available data. As mentioned above, in Japan (although the data are semi-annual), the share of RMB-invoiced transactions in trade with Asia (both exports and imports) has been published since the second half of 2011, and the share in global trade has been published since the second half of 2013.¹⁵ Furthermore, since the first half of 2023, the share of renminbi-invoiced transactions in trade with the PRC and Hong Kong, China has also been published. In the Republic of Korea, the Bank of Korea publishes currency settlement shares for trade with the rest of the world and for bilateral trade with the PRC. In 2023, the share of renminbi settlements in the Republic of Korea's total imports from around the world was 2.4%, and it was 10.4% in imports from the PRC; similarly, the share of renminbi settlements in exports was 1.7% of total exports and 8.0% of exports to the PRC. However, as shown in Tomizawa and Shimizu (2026), only a limited number of economies publish data on the share of invoice and settlement currencies, including the renminbi. Furthermore, even when such data are published, they often appear on a quarterly or semi-annual basis, and there are significant limitations on the period covered—for example, in Europe, only data from the 2020s onwards are publicly available.

In this analysis, the dataset by Boz et al. (2025, IMF), published in September 2025, added the renminbi share to the invoice currency shares data (broken down by US dollar, euro, and home currency) previously reported by the same authors in their 2020 paper. Among widely available data on invoice currency shares, this dataset has the broadest coverage, spanning 132 countries and regions and extending up to 2023. For example, the 2023 data show that the share of renminbi-invoiced transactions in the Russian Federation and Mongolia has grown to around 20%–30%.

In Tomizawa and Shimizu (2026), the authors analyzed the shares and trends of the renminbi as an invoice currency in Asian countries and major Western nations based on data published by central banks and ministries of finance. However, for countries such as the Russian Federation and Mongolia, no data had been released by their respective ministries of finance or central banks as of July 2025, when the paper was written. Table 19 compares the renminbi-denominated share figures disclosed by public institutions in each country with the shares reported by Boz et al. For example, in Malaysia, the data published by the central bank are ad hoc and limited in scope (only the renminbi share of total trade in 2023), whereas Boz et al.'s dataset provides continuous coverage from 2013 onward. No discrepancies were found with data from national central banks, ministries of finance, and other sources.

¹⁵ We believe that the enhancement of such data disclosure is of significant importance in measuring the extent of expansion in actual trade transactions and their correlations, against the backdrop of the December 2011 Japan–PRC Financial Cooperation Agreement and the June 2012 launch of direct yen–yuan exchange, which occurred amid the ongoing RMB internationalization at the time.

Table 19: Data Comparison Between Official Data and Boz et al. (2023)

				2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
Thailand	Official Data	Export	World	0.1	0.2	0.3	0.3	0.4	0.5	0.5	0.5	0.9	1.1		
		Import	World	0.4	0.7	0.7	0.9	1.1	1.5	1.7	1.9	3.4	3.8		
	Boz et al. (2025)	Export	World	0.1	0.2	0.3	0.3	0.4	0.5	0.5	0.5	0.9			
		Import	World	0.4	0.7	0.7	0.9	1.1	1.5	1.7	1.9	3.4			
				2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
Indonesia	Official Data	Export	World					0.6	1.0	0.9	1.1	2.1	3.5		
		Import	World					1.9	3.3	3.6	4.6	6.4	8.8		
	Boz et al. (2025)	Export	World	0.8	0.7	0.3	0.4	0.6	1.0	0.9	1.1	2.0			
		Import	World	0.2	0.6	0.6	0.9	1.9	3.3	3.6	4.6	6.4			
				2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
Malaysia	Official Data	Trade (Ex+Im)	World	2.9											
		Export	World	0.3	0.3	0.5	0.5	0.8	1.3	1.0		2.3			
	Boz et al. (2025)	Export	World	0.7	1.0	1.4	1.7	2.2	2.3	2.5		3.8			
		Import	World												
				2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
Australia	Official Data	Export	World	0.1	0.2	0.3	0.3	0.3	0.4	0.2	0.1	0.2	0.2		
		Import	World	0.5	0.6	0.6	0.7	0.9	1.1	1.3	1.4	1.3	1.4		
	Boz et al. (2025)			2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
		Export	World	0.1	0.3	0.3	0.3	0.4	0.3	0.2	0.1	0.2			
				0.6	0.6	0.6	0.8	1.0	1.2	1.3	1.4	1.4			
				2015	2016	2017	2018	2019 4Q	2020 4Q	2021 4Q	2022 4Q	2023 4Q	2024 4Q		
Taipei,China	Official Data	Export	World					2.6	2.4	2.2	1.7	1.6	1.4		
		Import	World					1.1	1.2	1.2	1.1	1.5	1.5		
	Boz et al. (2025)			2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
		Export	World					2.1	2.3	2.5	2.5	2.4	2.3	1.6	1.6
								0.8	0.9	1.0	1.1	1.2	1.2	1.1	1.4
				2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
France	Official Data	Export	Extra-EU									2.7	2.7	2.5	
		Import	Extra-EU									1.5	1.9	1.9	
	Boz et al. (2025)	Export	World	0.4	0.4	0.5	0.5	0.6	0.9	1.3	1.2	1.2			
		Import	World	0.1	0.2	0.2	0.3	0.4	0.4	0.5	0.5	0.7			

Source: Collected from the websites of national central banks, etc.; Boz et al. (2025, IMF)

Consequently, in the empirical analysis of this paper, we selected 70 economies from the Boz et al. (2025, IMF) dataset for which renminbi shares are available, focusing on the period from 2016¹⁶ to 2023—a period for which comprehensive data are available—and adopted the renminbi's share of each economy's invoicing currency for exports and imports to the rest of the world as the dependent variable. Note that in the Boz et al. (2025, IMF) dataset, for economies such as the United States, Ecuador (dollarized), Panama, and Timor-Leste, where the home currency is the US dollar, as well as economies like those in Europe where the home currency is the euro, the home currency field is left blank (zero).¹⁷

4.2 Empirical Analysis Model

In this study, we conduct an empirical analysis of country-level panel data covering the period 2016 to 2023 to examine the factors driving the growth of the renminbi-invoiced share of trade, while previous studies, such as Boz et al. (2025), utilized data on the currency shares of trade settlements in global exports and imports, and Perez-Saiz and Zhang (2023) calculated the renminbi share using SWIFT transaction data. However, as explained in the previous section, this paper uses the renminbi-denominated share data published by Boz et al. (2025) as the dependent variable and constructs

¹⁶ The year 2015 is used as a lag variable (as the value from one period prior to 2016).

¹⁷ Data for the PRC and Hong Kong, China are not included in the 132 economies and regions in the Boz et al. (2025) dataset.

a dataset that delves deeper into the explanatory variables related to the renminbi internationalization policies (currency swap lines and offshore renminbi settlement banks) implemented by the People's Bank of China (PBC) used in the analysis by Perez-Saiz and Zhang (2023).

In this paper, following prior studies such as Novy (2006), Goldberg and Tille (2008), and Ito and Chinn (2014), which analyzed the roles of fundamentals and policy in determining the choice of trade settlement currency, we employ the following regression model.

$$RMB_{i,t}^k = \alpha_0^k + \alpha_1^k \cdot RMB_{i,t-1}^k + \alpha_2^k \cdot MF_{i,t} + \alpha_3^k \cdot ER_{i,t} + \alpha_4^k \cdot China Policy_{i,t} + \varepsilon_{i,t}^k \quad (1)$$

The dependent variable, $RMB_{i,t}^k$, uses the renminbi-denominated share of country i 's exports and imports (k denotes exports and imports) at time t (data published by Boz et al. 2025).¹⁸ Since the share data are available only through 2023, the analysis period is set from 2015 to 2023, when various renminbi internationalization policies began to progress in earnest. The sample consists of 70 countries, as in Boz et al. (2025). However, since missing values occur when data for certain explanatory variables are unavailable, this constitutes an unbalanced panel analysis. Boz et al. (2025) used a time-only fixed-effects model, Perez-Saiz and Zhang (2023) employed fixed-effects and random-effects models (including pooled OLS), and Ito and Chinn (2014) used a random-effects model; therefore, this paper tests all of these models and summarizes the results.^{19, 20}

As an explanatory variable, we first use the renminbi-denominated share from the previous period. This is because inertia plays a role in the choice of invoice currency, as confirmed by Boz et al. (2025) and Perez-Saiz and Zhang (2023), and we adopted a model consistent with previous research.²¹ Next, the vector $MF_{i,t}$ contains data representing country i 's macro fundamentals, including country i 's GDP (log), the value of exports and imports with the PRC (log), and the trade share with the PRC (the ratio of the value of exports and imports with the PRC to country i 's total exports and imports). $ER_{i,t}$ contains data on foreign exchange and capital markets, including the rate of change in country i 's exchange rate against the renminbi; a foreign exchange anchor dummy variable that takes a value of 1 if country i 's exchange rate regime uses the US dollar as an anchor and 0 otherwise; an indicator of the degree of regulation in country i 's foreign exchange and capital markets; and finally, $China Policy_{i,t}$, which comprises various explanatory variables, including the

¹⁸ When the dependent variable is a share, using the fractional regression model with logit or probit proposed by Papke and Wooldridge (1996) allows us to appropriately handle the fact that the invoice currency share is constrained to the range of 0 to 1. However, it is well known that the so-called "random parameter problem" becomes more severe in nonlinear panel data models with fixed effects, and previous studies have adopted the invoice currency share as the dependent variable as is.

¹⁹ Ito and Chinn (2014) state that they adopted a random-effects model because they believe that each country is exposed to country-specific shocks that are not inherently intrinsic to the country itself. At the same time, recognizing the possibility that country-specific characteristics—such as the institutional and regulatory environment—may influence the choice of invoicing currency, they also conducted estimates using a fixed-effects model as a robustness check.

²⁰ We estimated the data using pooled OLS, a fixed-effects model, and a random-effects model. However, based on the results of the redundant fixed-effects test and the Hausman test, the random-effects model was rejected across all estimations, and the fixed-effects model was deemed appropriate in most cases.

²¹ We have confirmed that excluding the RMB-denominated share from the previous period does not significantly affect the signs or significance of the explanatory variables.

PRC's policy measures regarding the internationalization of the renminbi and the current status of such policies.

As regards the selection of explanatory variables, as detailed in the previous section, the basic model uses the KAOPEN index (2.28 to -1.94 ; a value closer to 2.28 indicates looser restrictions on capital transactions in that country) as an indicator of the degree of regulation in country i 's foreign exchange and capital markets. In the extended model, instead of the KAOPEN index, we counted five items related to regulations on local and foreign currencies (which are considered to have a more direct impact depending on whether the country's own currency or the counterparty's currency is selected as the invoice currency) and foreign exchange. We assigned a value of 1 if regulations were in place and 0 if they were not, creating a variable with a range from 0 to 5. A score of 5 indicates the strongest currency and foreign exchange regulations.

Next, regarding the state of financial infrastructure, in the basic model, a clearing dummy was set to 1 if a renminbi clearing bank was established in country i and to 0 otherwise. Furthermore, in the extended model (as an alternative to quantifying currency swap agreements), we quantified the number of Chinese banks operating locally. We scored the business scale of local subsidiaries of the PRC's five major commercial banks entering each country as 3, branches of the Chinese headquarters as 2, and representative offices as 1.

Furthermore, as a policy variable in the extended model and an indicator of political proximity, we added a variable calculated by summing the number of visits by PRC leaders (President, Premier) to partner countries and the number of visits by partner country leaders (President, Prime Minister) to the PRC since 2010, when the internationalization of the renminbi effectively began.

In terms of geographical proximity, we designated the 36 countries connected to the China–Europe Railway Express—a key transport infrastructure of the Belt and Road Initiative that links the PRC and Europe via the Silk Road—as countries eligible for the China–Europe Railway Express dummy variable, assigning a value of 1 to these countries and 0 to those not connected. Finally, as a digital-related variable, we assigned a value of 1 to the DX-related dummy variable for countries that have signed an e-commerce (EC) MOU with the PRC to promote EC trade, and 0 to countries that have not signed such an agreement.

4.3 Empirical Analysis Results: Share of RMB-invoiced Imports

Table 20 summarizes the results for the share of renminbi-invoiced imports, the dependent variable, with baseline explanatory variables. In estimates using the share of renminbi-invoiced imports as the dependent variable, the one-period lag term was positive in all models and statistically significant at the 1% level. This result, in line with prior research, indicates that renminbi-invoiced import transactions exhibit strong inertia and persistence.

The share of imports from the PRC was positive and significant in all specifications, confirming that countries with greater trade dependence on the PRC have a higher proportion of RMB-invoiced imports. On the other hand, while the value of imports from the PRC was not significant in the baseline model, it became significantly negative in models with expanded explanatory variables, suggesting that an expansion in the scale of imports does not necessarily lead to an increase in the use of renminbi-invoiced transactions.

Table 20: Share of RMB-Invoiced Import (Baseline Specification)

Dependent Variable: RMB-Invoiced Share of Imports								
Sample Period: 2016–2023								
Cross-section: 66								
Total panel (unbalanced) Observations: 524	Pooled OLS		Fixed Effects		Fixed Effects		Random Effect	
			Fixed-effects model with time-only effects		Fixed-effects model for country and period		Country-specific random-effects model	
Constant term	-0.0138	(0.2450)	0.0308	(0.2456)	4.1651 *	(2.3708)	-0.0039	(0.2907)
RMB-Invoiced share_imports(-1)	1.1405 ***	(0.0247)	1.1338 ***	(0.0248)	1.1874 ***	(0.0624)	1.1572 ***	(0.0281)
Share of imports from the PRC	0.0326 ***	(0.0079)	0.0333 ***	(0.0079)	0.2245 ***	(0.0315)	0.0386 ***	(0.0092)
Value of imports from the PRC (log)	0.0013	(0.0306)	-0.0060	(0.0307)	-0.6207 **	(0.2832)	-0.0017	(0.0361)
Exchange rate change (vs. RMB)	1.3384 ***	(0.4007)	1.7131 ***	(0.4251)	2.3072 ***	(0.4981)	1.4417 ***	(0.3952)
Exchange rate anchor (USD)	-0.4463 ***	(0.1279)	-0.4619 ***	(0.1280)	0.1393	(1.0349)	-0.5327 ***	(0.1507)
KAOPEN Index	-0.4296 ***	(0.1614)	-0.4029 **	(0.1612)	-2.2260 ***	(0.5795)	-0.5018 ***	(0.1874)
Clearing Bank Dummy	0.2397 **	(0.1113)	0.2332 **	(0.1110)	-0.2715	(0.4363)	0.2577 **	(0.1305)
Adjusted R-squared	0.8379		0.8389		0.8684		0.7892	
Redundant Fixed-Effects Tests			1.4768		2.7525/2.4582			
Correlated Random Effects/Hausman Test							65.6967	

Based on the author's estimates. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

In regard to exchange rate factors, the rate of change in the exchange rate against the renminbi (a positive value indicates that the partner country's currency is depreciating against the renminbi) was positive and highly significant in all models, indicating that depreciation of the partner country's currency—or, in other words, appreciation of the renminbi—has the effect of promoting the use of renminbi-invoiced imports. This is consistent with the empirical findings of a similar analysis by Shimizu et al. (2024) using Japanese customs data, which showed that yen depreciation negatively affects yen-invoiced transactions. On the other hand, the exchange rate anchored to the US dollar was consistently negative and significant, confirming a tendency for renminbi-invoiced imports to be suppressed in countries that adopt exchange rate regimes with a strong tendency to track the US dollar. This is also consistent with the results of previous studies.

With respect to institutional factors, the capital account openness index (KAOPEN) was found to be significantly negative, indicating that countries with relatively free capital mobility (where a higher KAOPEN value signifies greater freedom of capital movement) have a lower share of renminbi-invoiced imports. In other words, countries with remaining restrictions on capital mobility tend to utilize renminbi-invoiced transactions. Furthermore, consistent with the results of Boz et al. (2025), who estimated the same data using a nearly identical model, the clearing bank dummy variable was positively and significantly associated, confirming that the development of renminbi settlement infrastructure promotes renminbi-invoiced imports.

Next, in the model with expanded explanatory variables presented in this paper, an explanatory variable based solely on the number of foreign exchange transaction restrictions collected by the IMF's AREAER—used in place of KAOPEN, which covers general cross-border capital mobility regulations—yielded positive and significant results (Table 21). This indicates that countries imposing foreign exchange controls—such as the PRC—are beginning to use the renminbi despite both currencies being subject to such controls. Instead of the clearing bank dummy or the currency swap with the PRC dummy, which have traditionally been used as proxy variables for the PRC's RMB internationalization policy in previous studies, we have identified explanatory

variables that more realistically reflect RMB internationalization policy, including political actions: the number of branches established by the five major Chinese banks in various countries, visits by PRC dignitaries and by leaders of partner countries to the PRC, a China–Europe Railway Express dummy variable indicating the development of infrastructure connecting the PRC and European countries by land, and a DX-related MOU dummy variable (a dummy variable set to 1 in the year of signing). As a result, positive and statistically significant results were obtained for all variables except the number of branches of Chinese banks. These results suggest that the strengthening of institutional, diplomatic, and logistical ties contributes to the expansion of renminbi-invoiced imports.

Table 21: Share of RMB-Invoiced Import (Extended Specification)

Dependent Variable: RMB-Invoiced Share of Imports						
Sample Period: 2016–2023						
Cross-section: 69						
Total panel (unbalanced) Observations: 547	Pooled OLS		Fixed Effects		Random Effect	
			Fixed-effects model with time-only effects		Country-specific random-effects model	
Constant term	0.0096	(0.2665)	0.1023	(0.2688)	-0.0701	(0.3052)
RMB-Invoiced share_imports(-1)	1.1220 ***	(0.0245)	1.1137 ***	(0.0247)	1.1364 ***	(0.0272)
Share of imports from the PRC	0.0344 ***	(0.0076)	0.0356 ***	(0.0076)	0.0387 ***	(0.0086)
Value of imports from the PRC (log)	-0.0627 *	(0.0377)	-0.0750 **	(0.0380)	-0.0592	(0.0431)
Exchange rate change (vs. RMB)	1.3292 ***	(0.3917)	1.7201 ***	(0.4152)	1.4071 ***	(0.3854)
Exchange rate anchor (USD)	-0.4527 ***	(0.1221)	-0.4707 ***	(0.1221)	-0.5272 ***	(0.1392)
Number of exchange controls	0.0600 **	(0.0293)	0.0580 **	(0.0292)	0.0750 **	(0.0335)
Number of Bank of China branches	0.0069	(0.0144)	0.0091	(0.0144)	0.0013	(0.0165)
Number of visits by Chinese officials	0.2186 ***	(0.0736)	0.2264 ***	(0.0734)	0.2423 ***	(0.0845)
China-Europe Railway Express dummy	0.2624 **	(0.1092)	0.2699 **	(0.1091)	0.2681 **	(0.1263)
DX-related MOUs (dummy)	0.3217 **	(0.1313)	0.2852 **	(0.1324)	0.3268 **	(0.1410)
Adjusted R-squared	0.8395		0.8408		0.7968	
Redundant Fixed-Effects Tests			1.6343			
Correlated Random Effects/Hausman Test					71.2356	

Based on the author's estimates. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

4.4 Empirical Analysis Results: Share of RMB-Invoiced Exports

In analyses in which the share of renminbi-invoiced exports was the dependent variable, the one-period lag term was positive and statistically significant at the 1% level across all models, confirming that renminbi-invoiced transactions exhibit extremely high persistence in exports as well.

Table 22 summarizes the results for the share of renminbi-invoiced exports, the dependent variable, with baseline explanatory variables. The share of exports to the PRC was positive and statistically significant in many models, indicating that countries with greater dependence on the Chinese market tend to engage more in renminbi-invoiced exports. On the other hand, the **logarithm of export value to the PRC** was significantly negative in some models, suggesting that an expansion in export scale does not necessarily promote the use of renminbi-invoiced exports.

Table 22: Share of RMB-Invoiced Export (Baseline Specification)

Dependent Variable: RMB-Invoiced Share of Exports								
Sample Period: 2016–2023								
Cross-section: 66								
Total panel (unbalanced) Observations: 524	Pooled OLS		Fixed Effects		Fixed Effects		Random Effect	
			Fixed-effects model with time-only effects		Fixed-effects model for country and period		Country-specific random-effects model	
Constant term	0.2072	(0.1760)	0.2178	(0.1750)	0.4066 *	(1.1453)	0.2259	(0.1841)
RMB-Invoiced share of exports (-1)	1.1362 ***	(0.0354)	1.1255 ***	(0.0354)	1.0721 ***	(0.0637)	1.1324 ***	(0.0362)
Share of Exports to the PRC	0.0131 *	(0.0069)	0.0151 **	(0.0069)	0.1092 ***	(0.0229)	0.0144 **	(0.0071)
Export value to the PRC (log)	0.0147	(0.0211)	0.0104	(0.0210)	0.0628	(0.1338)	0.0143	(0.0220)
Exchange rate change (vs. RMB)	0.1539	(0.4027)	0.5334	(0.4258)	1.1751 **	(0.5212)	0.1732	(0.3993)
Exchange rate anchor (USD)	-0.2261 *	(0.1240)	-0.2547 **	(0.1236)	-0.0805	(1.0934)	-0.2435 **	(0.1295)
KAOPEN Index	-0.4483 ***	(0.1633)	-0.4199 **	(0.1625)	-2.1745 ***	(0.6113)	-0.4747 ***	(0.1701)
Clearing Bank Dummy	0.1126	(0.1104)	0.1028	(0.1098)	-0.2660	(0.4572)	0.1133	(0.1153)
Adjusted R-squared	0.8501		0.8521		0.8648		0.8332	
Redundant Fixed-Effects Tests			1.9791		1.7373/3.1256			
Correlated Random Effects/Hausman Test							33.6372	

Based on the author's estimates. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

With regard to the exchange rate change, while the significance is weaker than for imports, it is sometimes positive and significant in the extended model. The exchange rate anchored to the US dollar is significantly negative in many estimates, confirming that a dollar-centric exchange rate regime also suppresses the use of renminbi-invoiced exports.

Furthermore, capital account openness (KAOPEN) is consistently negative and significant, and the results show that the higher a country's degree of financial freedom, the lower the proportion of renminbi-invoiced exports. On the other hand, the clearing bank dummy is often not significant for exports, suggesting that the effect of settlement infrastructure may be relatively limited compared to imports.

In the extended specification (Table 23), the (log) value of exports to the PRC was negative and significant in some models (−0.04 to −0.045). As with imports, this suggests that an expansion in the scale (value) of exports does not necessarily lead to an increase in the renminbi-denominated ratio. Among Chinese policy variables, currency and exchange controls were positive and significant (0.050 to 0.054). These results suggest that countries with more currency and exchange controls (i.e., lower openness) may have a higher renminbi-invoiced ratio. As with imports, this implies that countries like the PRC, which impose restrictions on currency and exchange transactions, are beginning to use the renminbi despite both currencies being subject to controls.

As with imports, the results for exports showed a positive correlation with the number of Chinese-owned banks entering the market under both models, though this was not statistically significant. Positive and statistically significant results were obtained for the other Chinese policy variables. In regard to the number of high-level visits, the results were positive and statistically significant (0.241–0.247), as was the case with imports. This suggests that political and geopolitical proximity, manifested in mutual visits and strengthened relations between the leaders of the PRC and the partner country, leads to increased renminbi usage. As regards the China–Europe Railway Express and the dummy variable, as with imports, the results were positive and significant (0.229–0.235), which suggests that strengthening critical infrastructure and logistics

connectivity could expand renminbi usage. Similarly to imports, the DX-related MOU dummy variable also yielded a positive and significant result (0.291–0.330), suggesting that cooperation frameworks in the digital sector and the promotion of e-commerce transactions may expand renminbi usage.

Table 23: Share of RMB-Invoiced Export (Extended Specification)

Dependent Variable: RMB-Invoiced Share of Exports					
Sample Period: 2016–2023					
Cross-section: 69					
Total panel (unbalanced) Observations: 547	Pooled OLS		Fixed Effects		Random Effect
			Fixed-effects model with time-only effects		Country-specific random-effects model
Constant term	0.0364	(0.1550)	0.0728	(0.1542)	0.0356 (0.1535)
RMB-Invoiced share of exports (-1)	1.0964 ***	(0.0367)	1.0832 ***	(0.0368)	1.0957 *** (0.0362)
Share of Exports to the PRC	0.0237 ***	(0.0076)	0.0265 ***	(0.0076)	0.0239 *** (0.0075)
Export value to the PRC (log)	-0.0463 *	(0.0246)	-0.0517 **	(0.0244)	-0.0465 * (0.0243)
Exchange rate change (vs. RMB)	0.1601	(0.3888)	0.5790	(0.4117)	0.1630 (0.3813)
Exchange rate anchor (USD)	-0.2749 **	(0.1153)	-0.3126 ***	(0.1151)	-0.2779 ** (0.1142)
Number of exchange controls	0.0566 *	(0.0291)	0.0542 *	(0.0289)	0.0576 ** (0.0288)
Number of Bank of China branches	-0.0029	(0.0138)	-0.0036	(0.0137)	-0.0032 (0.0137)
Number of visits by Chinese officials	0.1512 ***	(0.0436)	0.1573 ***	(0.0434)	0.1522 *** (0.0432)
China-Europe Railway Express dummy	0.2430 **	(0.1077)	0.2472 **	(0.1071)	0.2443 ** (0.1068)
DX-related MOUs (dummy)	0.3390 ***	(0.1303)	0.3014 **	(0.1318)	0.3391 ** (0.1286)
Adjusted R-squared	0.8540		0.8561		0.8506
Redundant Fixed-Effects Tests			2.1333		
Correlated Random Effects/Hausman Test					34.8141

Based on the author's estimates. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

4.5 Discussion

Based on the above empirical results, it has become clear that the share of renminbi-invoiced trade exhibits strong persistence and is systematically influenced by trade dependence on the PRC, exchange rate regimes, capital controls, settlement infrastructure, and strengthened political, institutional, and logistical ties. In particular, it was consistently confirmed for both imports and exports that while a US dollar-oriented exchange rate regime and the liberalization of capital movements tend to suppress the use of renminbi-invoiced transactions, renminbi-invoiced trade is promoted in emerging and developing economies where exchange controls remain in place, where the local currency tends to depreciate against the renminbi, where the PRC has established a railway infrastructure network, and where political ties with the PRC are strong. The empirical results of this study indicate that the expansion of RMB-invoiced trade does not proceed automatically through an increase in trade scale alone but is determined by the persistence of practices, the institutional environment, the exchange rate regime, and qualitative aspects of relations with the PRC. While this finding is generally consistent with existing theories explaining the use of international currencies, it carries several important implications.

(1) Currency Internationalization and the Role of “Inertia”

The fact that the one-period lag term of the renminbi-invoiced share was strongly and significantly positive for both imports and exports is consistent with discussions of ****network externalities and inertia**** regarding the use of international currencies. Existing research has pointed out that once the use of an international currency becomes established, it persists in a self-reinforcing manner through reduced transaction costs, learning effects, and the solidification of contractual practices. The results of this study suggest that, even though the renminbi remains a limited international currency compared to the dollar, a similar inertia can be observed even at the stage of partial internationalization. This can be interpreted as supporting the view that RMB internationalization is not a “phenomenon that progresses all at once” but rather a process that gradually takes root in specific countries and trade relationships.

(2) Trade Dependency and Choice of Transaction Currency

The finding that a country’s share of trade with the PRC consistently promotes RMB-invoiced transactions is consistent with theories regarding dominant trading partners. International trade theory posits that a trading partner’s market size and bargaining power influence the choice of contract currency, and the results of this study indicate that RMB-invoiced settlements are more readily accepted when the PRC is a major trading partner for a specific country.

On the other hand, the finding that the volume (scale) of trade itself does not necessarily have a positive effect is significant. This implies that renminbi-invoiced transactions do not expand naturally due to “large-scale trade” but are instead selectively employed within specific institutional and policy frameworks, which suggests that the use of international currencies depends heavily not only on market size but also on institutional design and policy guidance.

(3) Exchange Rate Regimes and Currency Competition

The finding that the exchange rate peg to the US dollar suppresses RMB-invoiced transactions in both imports and exports can be understood from the perspective of ****currency competition****. In countries where the exchange rate system is designed around the dollar, foreign exchange risk management, financial products, and settlement infrastructure are established on the premise of dollar denomination, making the switching costs to renminbi-invoiced transactions high.

This outcome indicates that the internationalization of the renminbi is not proceeding as a comprehensive replacement for the US dollar but is constrained by differences in exchange rate regimes and financial structures. In other words, the renminbi is not a “complete substitute” for the dollar; rather, its use is likely expanding as a complementary currency within specific institutional environments.

(4) Capital Controls and Institutional Trust

The fact that the capital account openness (KAOPEN) was negative and statistically significant appears, at first glance, to contradict currency internationalization theory. Conventional wisdom holds that the liberalization of capital movements promotes the international use of the currency; however, the results of this study suggest that capital controls may be supporting renminbi-invoiced trade.

This is consistent with the PRC’s policy strategy of “advancing renminbi internationalization in stages, starting with trade settlements, prior to capital liberalization.” It can be interpreted that countries with freer capital mobility tend to

prefer existing international currencies such as the dollar or the euro, thereby reducing the relative appeal of the renminbi, whereas within a certain regulatory environment, the renminbi is more easily integrated into institutional frameworks.

(5) The Role of Political, Institutional, and Logistical Factors

The consistently and significantly positive effects of variables such as visits by PRC government officials, the China–Europe Railway Express, and DX-related MOUs indicate that renminbi-invoiced trade is supported not only by market mechanisms but also by the deepening of political and institutional relationships.

This implies a shift from the conventional framework, which views the use of international currencies as a “neutral outcome of spontaneous market choice,” toward one that emphasizes state-led and relationship-dependent aspects. In the case of the renminbi, it is following an internationalization path distinct from that of typical reserve currencies, in that infrastructure development, diplomatic relations, and policy coordination drive currency usage.

(6) Asymmetry Between Imports and Exports

Finally, the fact that the significance of certain factors differs between imports and exports suggests that RMB internationalization depends primarily on the PRC’s policy intentions and trade structures. In particular, the stronger influence of settlement infrastructure and institutional factors on imports indicates that the renminbi may have first gained traction as a “currency for payments from the PRC.”

In summary, the findings reveal that while RMB internationalization is still in its early stages, it is a gradual process with strong inertia. It depends heavily not only on the scale of trade with the PRC but also on institutional and relational factors and is expanding in a complementary manner within the dollar-centered international monetary system. This presents a more limited and institution-dependent picture of RMB internationalization, in contrast to the view that characterizes it as the “rise of a new reserve currency to replace the dollar.”

5. CONCLUSION AND FUTURE RESEARCH

In this paper, we organize the variables used in prior research on the choice of invoice currency in trade. We then cross-reference these with the results of analyses of the mechanisms that promote local currency transactions and the factors that may facilitate them. We meticulously extracted new candidate variable data from various statistics and publicly available information and conducted an empirical analysis. In this paper, we used data on renminbi-invoiced shares published by Boz et al. (2025), and we constructed and analyzed a dataset that delves deeper into the explanatory variables related to the renminbi internationalization policies implemented by the People’s Bank of China (PBC)—such as currency swap lines and dummy variables for offshore renminbi settlement banks—which were used in the analysis by Perez-Saiz and Zhang (2023), by framing them as the activities of Chinese banks operating abroad.

In addition to the macroeconomic factors used in previous studies, the authors focused on capital control variables related to currencies where regulations are considered to have a stronger association with local currency transactions, and constructed variables based on these. Furthermore, to refine the geopolitical distance indicators (such as the distance between capitals and UN voting behavior dummies) presented in previous studies, the analysis counted and analyzed connections via freight railways—a critical

infrastructure—and high-level exchanges among leaders. Additionally, with regard to digital factors (not used in previous studies), the study organized and used the existence and duration of bilateral e-commerce agreements as variables.

The results reveal that although the internationalization of the renminbi is still in its early stages, it is a gradual process with strong inertia; it depends heavily on institutional and relational factors rather than on trade scale and is expanding in a complementary manner within the dollar-centered international monetary system. In other words, based on the experience of RMB internationalization to date, we have revealed that a combination of factors—particularly strong government initiatives, geopolitical elements, and digital factors—has led to the use of the renminbi in trade and investment. Furthermore, the finding that the renminbi is being chosen for transactions between countries with strict foreign exchange controls suggests that it is possible to increase the use of local currencies even in closed environments where currency and capital controls remain in place, which this paper considers a contribution.

On the other hand, it is important to note that the PRC's experience may not apply to all developing countries. To avoid falling into the so-called “middle-income trap,” developing countries must strive for stable macroeconomic management. At the same time, they should implement orderly currency and capital controls in step with the country's economic growth rate when it is experiencing momentum. In the medium term, this will lead to exchange rate stability, enhanced capital-raising capacity, and corporate growth, thereby strengthening confidence in the domestic currency. Even in developed countries like Japan, where there is limited room for deregulation, steps could be taken to deepen domestic financial markets and enhance their appeal. By seizing “opportunities” such as the digitization of currency and the movement of people, and by strengthening cooperation and collaboration with emerging economies, it may be possible to increase yen-denominated transactions within the Asian region and enhance the yen's presence as a regional currency.

One of the remaining issues identified through this study is that, while previous research found that a dummy variable for currency swaps with the PRC had a significant and positive effect on renminbi-invoiced transactions, using the actual swap amount rather than a dummy variable did not yield a significant result. Although we attempted to address this by comparing the swap amount to the country's import value, the result remained insignificant. Since it is believed that swap agreements with the PRC contribute to the expansion of RMB usage in practice, further refinement of the explanatory variables will likely be necessary. Second, there is the relationship between capital transactions—such as outward and inward direct investment—and the choice of currency for trade settlement. We had expected to find that in countries engaging in outward and inward direct investment with the PRC, the occurrence of RMB payments and receipts would lead to increased use of the renminbi for trade settlement, but this was not the case. We analyzed and identified the mechanisms that promote local currency–renminbi transactions in the PRC; however, due to capital (investment) flows with regions unique to the PRC, such as Hong Kong, China and Macau, China, we were unable to clarify the relationship between local currency-denominated transactions and capital transactions. The relationship between the establishment of capital-circulation routes to overseas markets through the relaxation of capital controls and the use of the local currency remains a topic for further research. Furthermore, Africa—which could not be examined in this study due to data limitations—is a region with which the PRC is building relationships and is a potential candidate for inclusion in the renminbi zone.

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